

CHAPTER 3: KINEMATICS IN TWO- DIMENSIONS

Subject: PNAP0123 Physics 1
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Topic 3: Kinematics in Two Dimensions

Understandings:

- Vectors and Scalars
- Mathematical Operations of Vectors: Graphical Method & Component Method
- Projectile Motion
- Solving Projectile Motion Problems
- Relative Velocity

Learning Outcomes

At the end of this chapter, students should be able to:

- Define Scalar And Vector Quantities.
- Perform Addition & Subtraction Of Vectors Using Graphical Method & Component Method.
- Resolve Vector Into Two Perpendicular Components (X and Y Axes).
- State The Properties In The Two-dimensional Motion Of Projectile Motion
- Solve Problems On Projectile Motion by applying Kinematics Equation
- Express Relative Velocity Of Object With Respect To A Reference Frame

3-1 Vectors and scalars

Vector and scalar quantities

- A **vector** quantity is one which has a magnitude (size) *and* a direction.
- A **scalar** quantity has only magnitude (size).

EXAMPLE: A force is a push or a pull and is measured in newtons. Explain why it is a vector.

SOLUTION: Suppose Joe is pushing Bob with a force of 100 newtons to the north.

- Then the magnitude of the force is 100 N.
- The direction of the force is north.
- Since the force has both magnitude and direction, it is a vector.

3-1 Vectors and scalars

Vector and scalar quantities

- A **vector** quantity is one which has a magnitude (size) *and* a direction.
- A **scalar** quantity has only magnitude (size).

EXAMPLE: Explain why time is a scalar.

SOLUTION: Suppose Joe times a foot race and the winner took 45 minutes to complete the race.

- The magnitude of the time is 45 minutes.
- But there is no direction associated with Joe's stopwatch. The outcome is the same whether Joe's watch is facing west or east. Time lacks any spatial direction. Thus time is a scalar.

3-1 Vectors and scalars

Vector and scalar quantities

- A **vector** quantity is one which has a magnitude (size) *and* a spatial direction.
- A **scalar** quantity has only magnitude (size).

EXAMPLE: Give examples of scalars in physics.

SOLUTION:

- Speed, distance, time, and mass are scalars.

EXAMPLE: Give examples of vectors in physics.

SOLUTION:

- Velocity, displacement, force, weight and acceleration are all vectors.

3-1 Vectors and scalars

Vector and scalar quantities

- Speed and velocity are examples of scalar and vector you are already familiar with.
- Speed is what your speedometer reads (say 35 km h^{-1}) while you are in your car. It does not care what direction you are going. Speed is a scalar.
- Velocity is a speed in a particular direction (say 35 km h^{-1} to the north). Velocity is a vector.

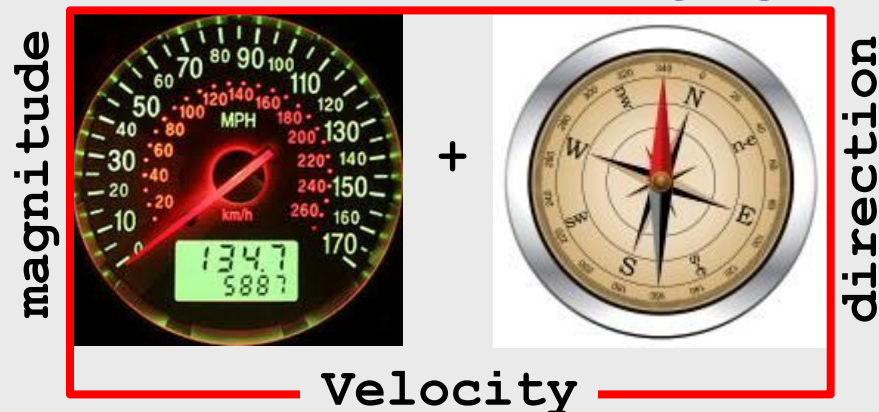
SCALAR



Speed

magnitude

VECTOR



Velocity

3-2 Addition of Vectors

Vector and scalar quantities

- Suppose the following movement of a ball takes place in 5 seconds.



- Note that it traveled to the right for a total of 15 meters in 5 seconds. We say that the ball's velocity is **+3 m/s** ($+15 \text{ m} / 5 \text{ s}$). The (+) sign signifies it moved in the **positive** x-direction.

- Now consider the following motion that takes 4 seconds.

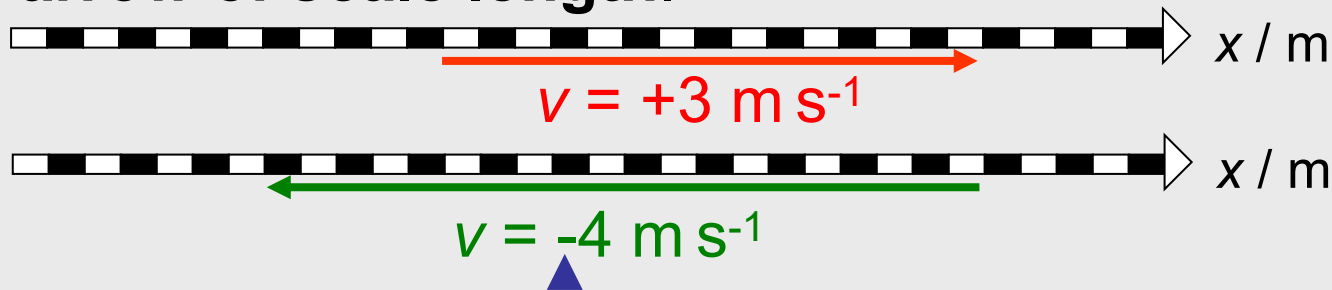


- Note that it traveled to the left for a total of 20 meters. In 4 seconds. We say that the ball's velocity is **-5 m/s** ($-20 \text{ m} / 4 \text{ s}$). The (−) sign signifies it moved in the **negative** x-direction.

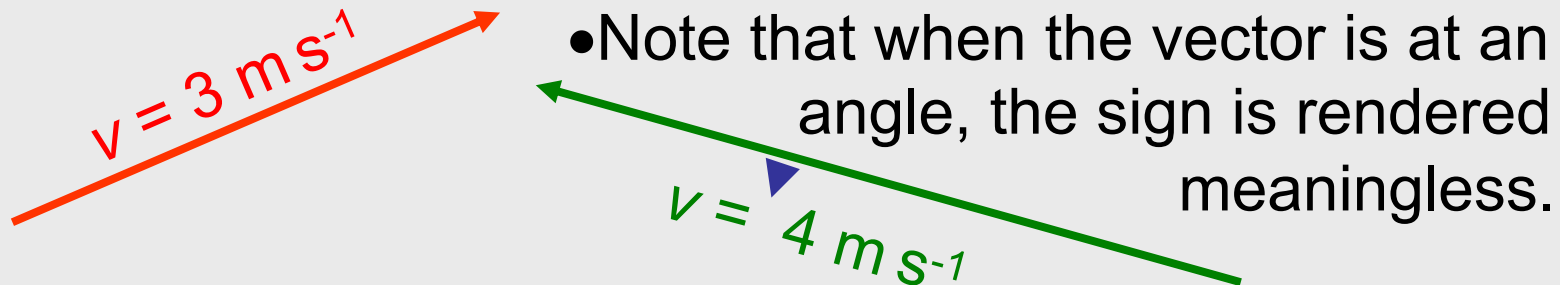
3-2 Addition of Vectors

Vector and scalar quantities

- It should be apparent that we can represent a vector as an **arrow of scale length**.



- There is no “requirement” that a vector must lie on either the x- or the y-axis. Indeed, a vector can point in any direction.



- Note that when the vector is at an angle, the sign is rendered meaningless.

3-2 Addition of Vectors- Graphical Methods

Combination and resolution of vectors

- We use simple arithmetic for adding scalars. Simple arithmetic can also be used for adding vectors if they are in the same direction.

- Distance for a person walks:

Addition Operation:

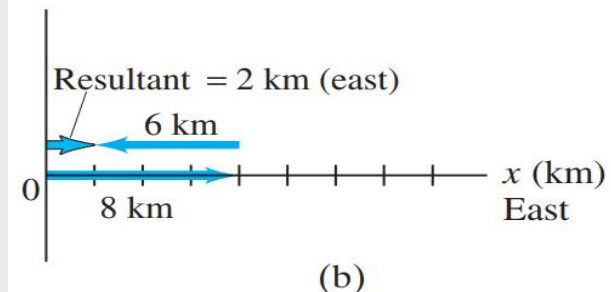
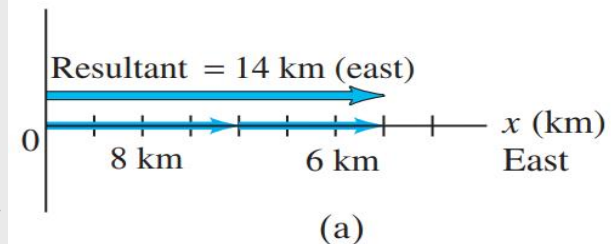
$$8 \text{ km} + 6 \text{ km} = 14 \text{ km, east}$$

Subtraction Operation:

$$8 \text{ km} - 6 \text{ km} = 2 \text{ km, east}$$

- However, simple arithmetic cannot be used if the two vectors are not along the same line.

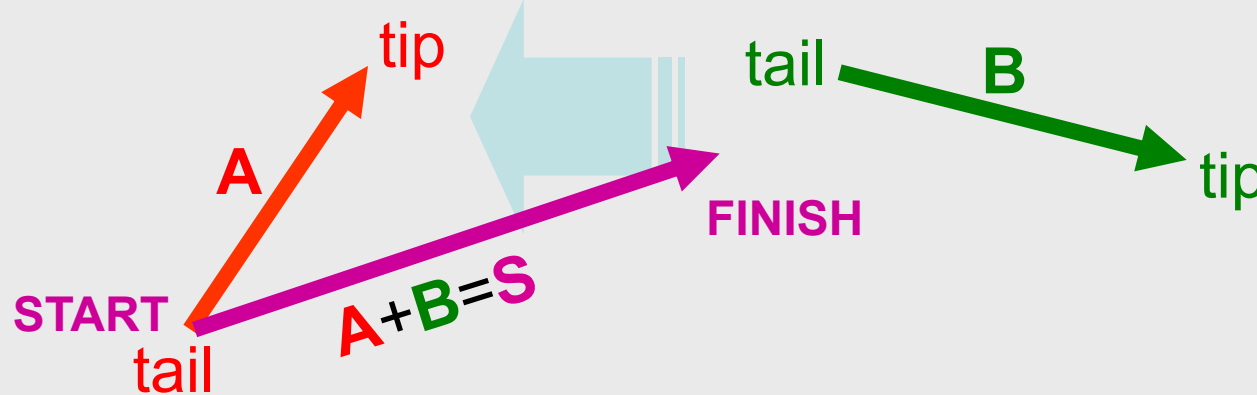
FIGURE 3-2 Combining vectors in one dimension.



3-2 Addition of Vectors—Graphical Methods

Combination and resolution of vectors (tail-to-tip)

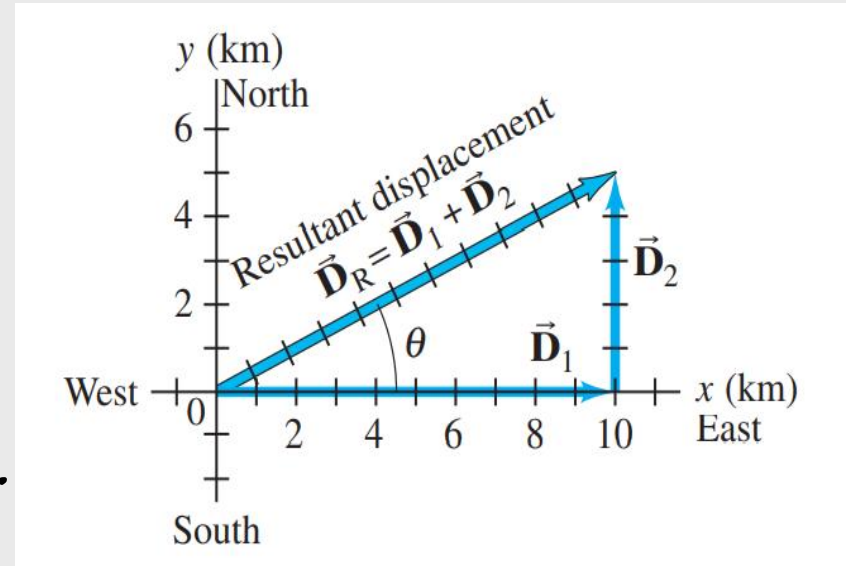
- Suppose we want to find the sum of the two vectors **A** + **B**.
- We take the second-named vector **B**, and **translate** it towards the first-named vector **A**, so that **B**'s TAIL connects to **A**'s TIP.
- The result of the sum, which we are calling the vector **S** (for sum), is gotten by drawing an arrow from the START of **A** to the FINISH of **B**.



3-2 Addition of Vectors- —Graphical Methods

Combination and resolution of vectors (tail-to-tip)

- E.g.: Suppose a person walks 10.0 km east and then walks 5.0 km north.
- Here, the travel paths are at right angles to one another, so we can find the displacement by using the Pythagorean Theorem.



- You can use the Pythagorean theorem only when the vectors are perpendicular to each other

$$D_R = \sqrt{D_1^2 + D_2^2} = \sqrt{(10.0 \text{ km})^2 + (5.0 \text{ km})^2} = \sqrt{125 \text{ km}^2} = 11.2 \text{ km.}$$

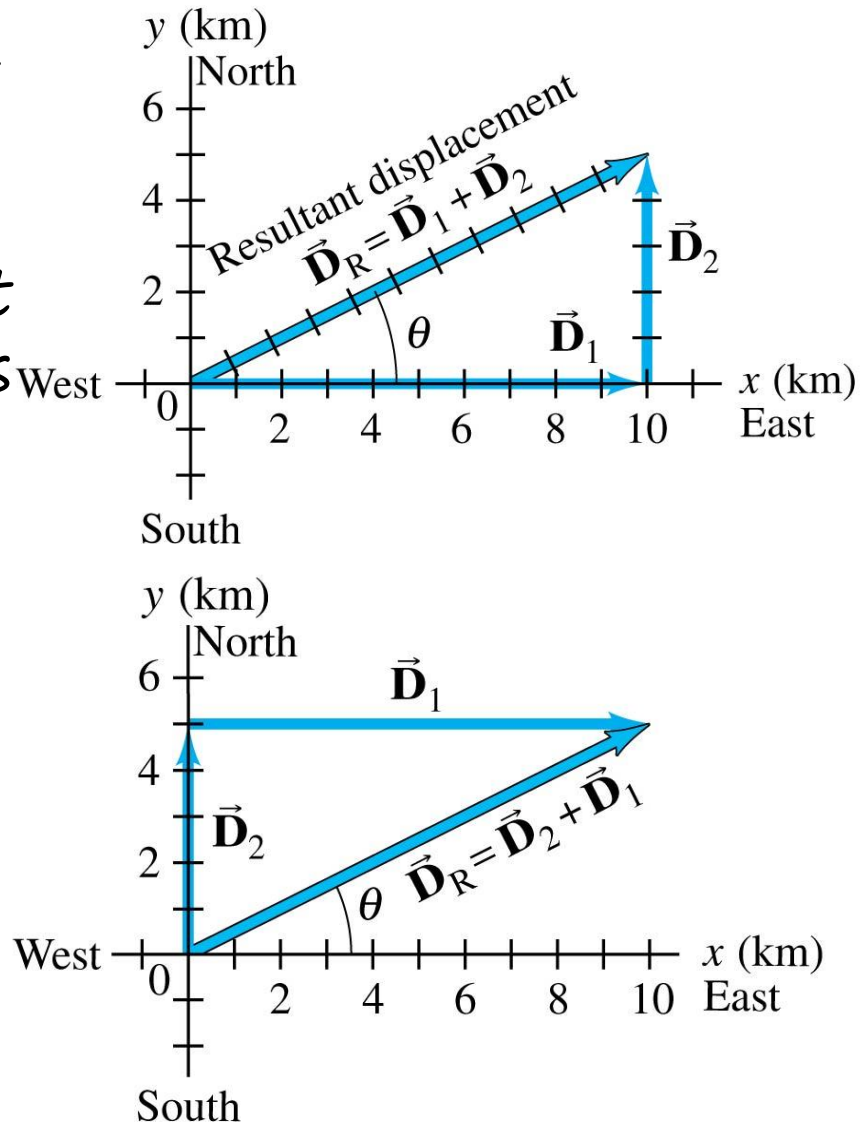
3-2 Addition of Vectors—Graphical Methods

- The resultant is not affected by the order in which the vectors are added.
- For example, a displacement of 5.0 km north, to which is added a displacement of 10.0 km east, yields a same resultant.
- Generally,

$$\vec{V}_1 + \vec{V}_2 = \vec{V}_2 + \vec{V}_1.$$

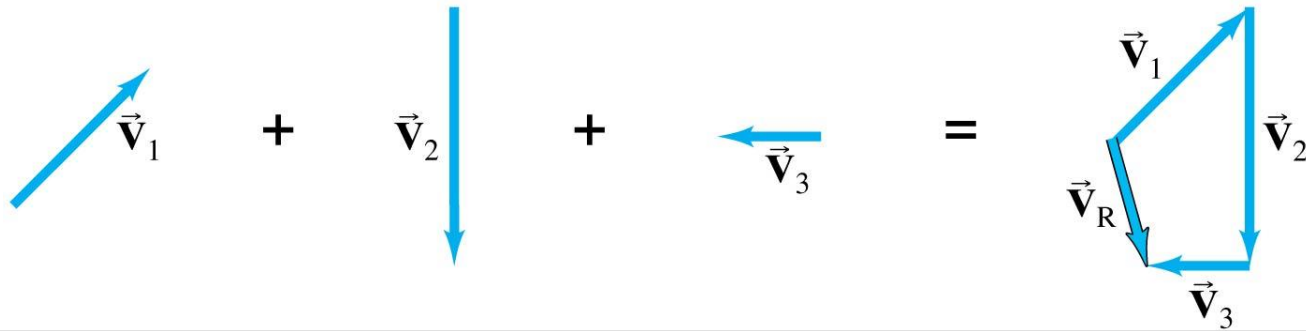
[As long as follow the rule of Tail-To-Tip adding method]

This is how we write ---> $\vec{D}_R = (11.2 \text{ km}, 27^\circ \text{ N of E})$

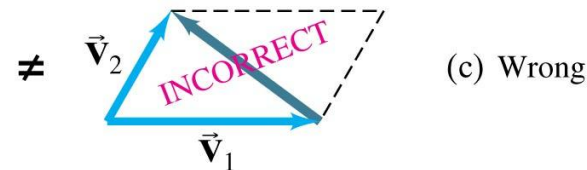
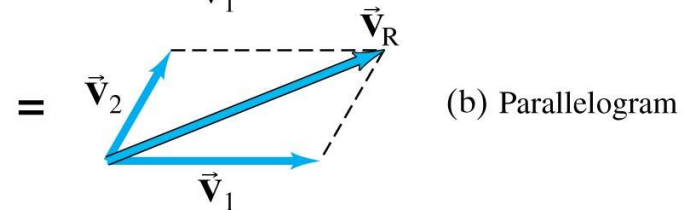
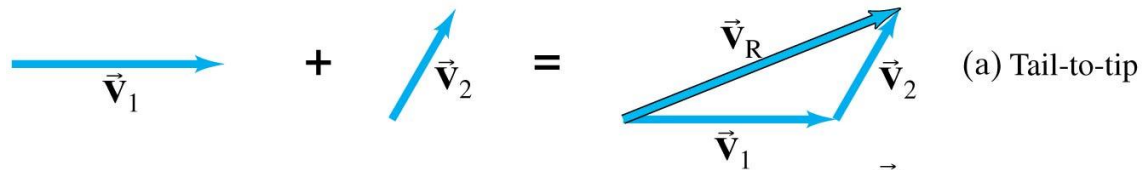


3-2 Addition of Vectors—Graphical Methods

Even if the vectors are not at right angles, they can be added graphically by using the “tail-to-tip” method.



The parallelogram method may also be used; “It is equivalent to tail-to-tip method”



Caution!

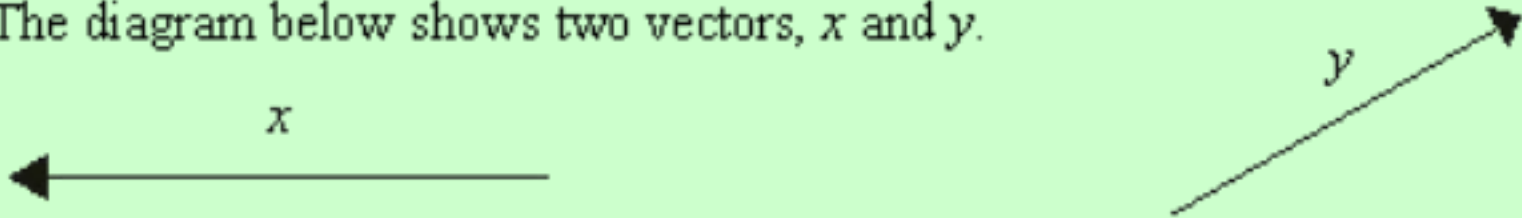
Be sure to use the correct diagonal on the parallelogram to get the resultant

(Measurement can be done on the graph paper with ruler and protractor to obtain the magnitude and the angle of the resultant vector.)

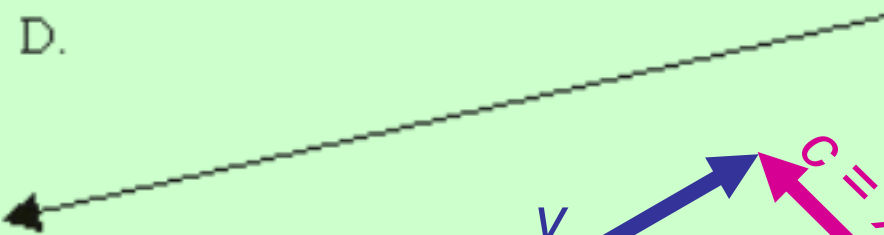
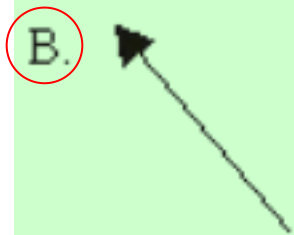
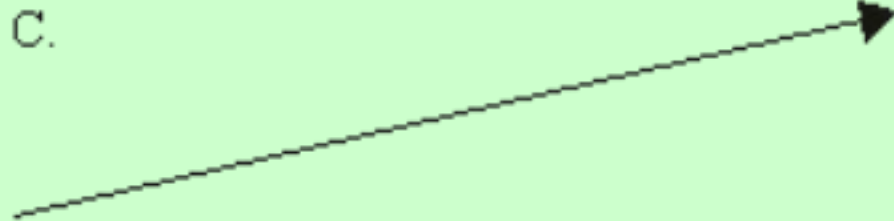
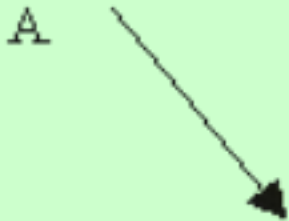
3-2 Addition of Vectors—Graphical Methods

Combination and resolution of vectors

The diagram below shows two vectors, x and y .

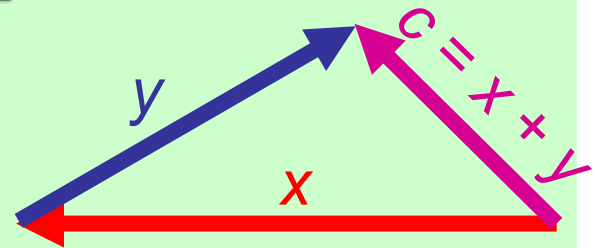


Which of the vectors below best represents the vector c that would satisfy the relation $c = x + y$?



SOLUTION:

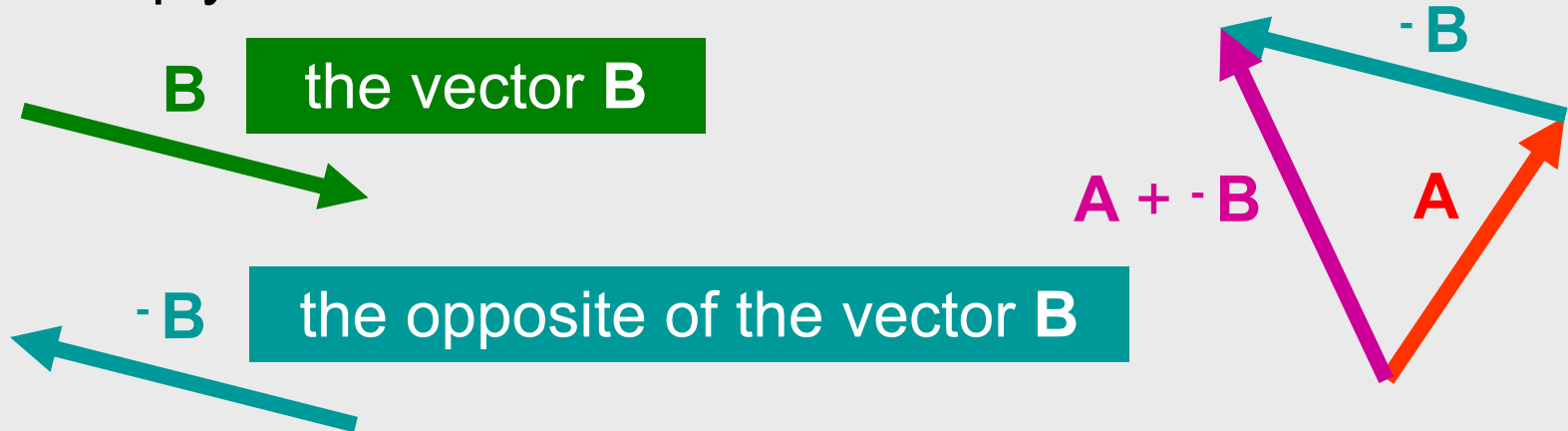
- Sketch the sum.



3-3 Subtraction of Vectors, and Multiplication of a Vector by a Scalar

Combination and resolution of vectors

- Just as in algebra we learn that to subtract is the same as to add the opposite ($5 - 8 = 5 + -8$), we do the same with vectors.
- Thus $\mathbf{A} - \mathbf{B}$ is the same as $\mathbf{A} + -\mathbf{B}$.
- All we have to do is know that the opposite of a vector is simply that same vector with its direction reversed.



Thus, $\mathbf{A} - \mathbf{B} = \mathbf{A} + -\mathbf{B}$

3-3 Subtraction of Vectors, and Multiplication of a Vector by a Scalar

Combination and resolution of vectors

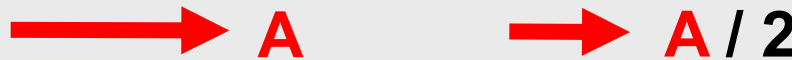
- To multiply a vector by a scalar, increase its length in proportion to the scalar multiplier.

- Thus, if **A** has a length of 3 m, then **2A** has a length of 6 m.

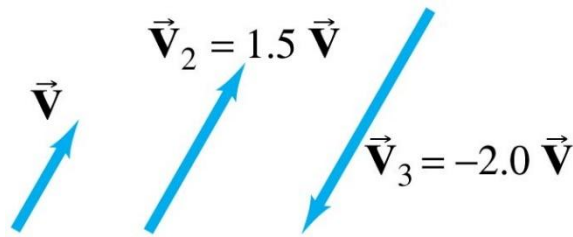


- To divide a vector by a scalar, simply multiply by the reciprocal of the scalar.

- Thus if **A** has a length of 3 m, then **A / 2** has a length of $(1/2)\mathbf{A}$, or 1.5 m.



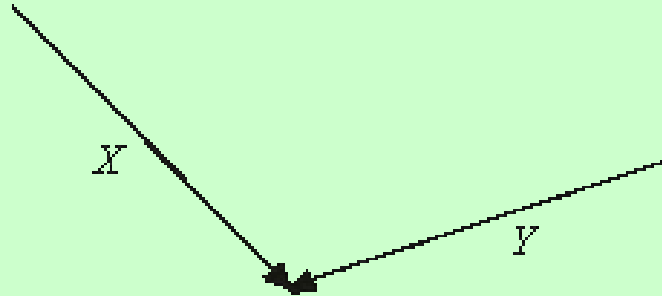
A vector $\vec{V}$ can be multiplied by a scalar c ; the result is a vector $c\vec{V}$ that has the same direction but a magnitude cV . If c is negative, the resultant vector points in the opposite direction.



3-3 Subtraction of Vectors, and Multiplication of a Vector by a Scalar

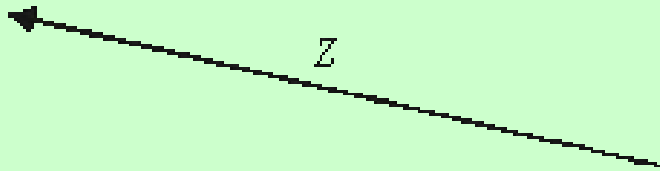
Combination and resolution of vectors

The diagram below shows two vectors X and Y .



Which of the following best represents the vector $Z = X - Y$.

A.



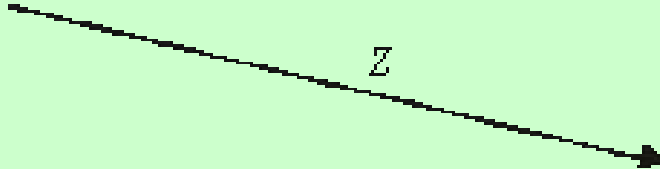
C.



D.

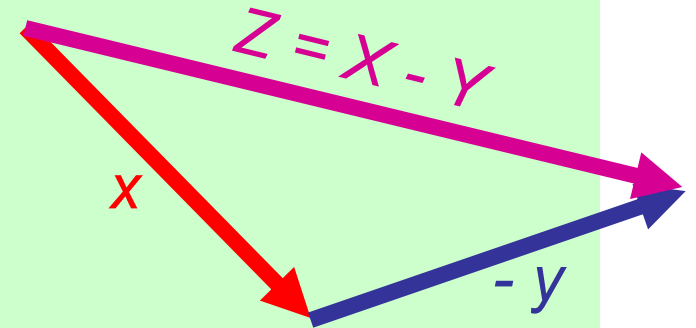


B.



SOLUTION:

- Sketch in the difference.



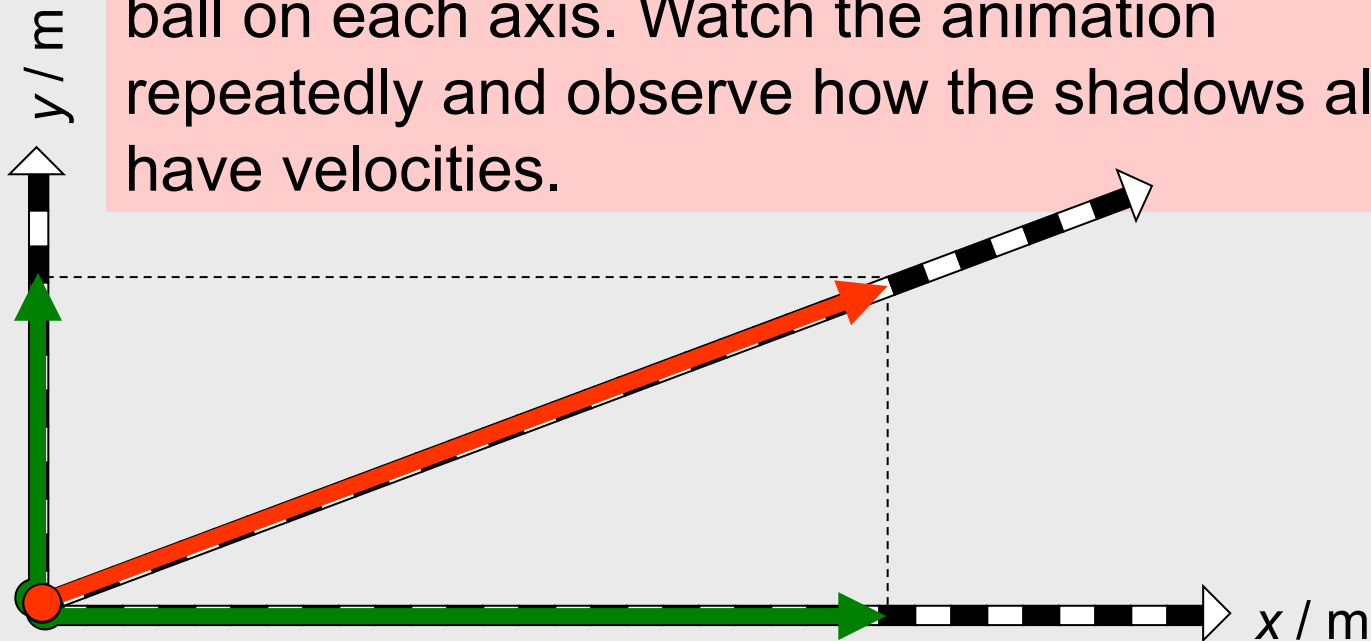
3-4 Adding Vectors by Components

Combination and resolution of vectors

- Suppose we have a ball moving simultaneously in the x- and the y-direction along the diagonal as shown:

FYI

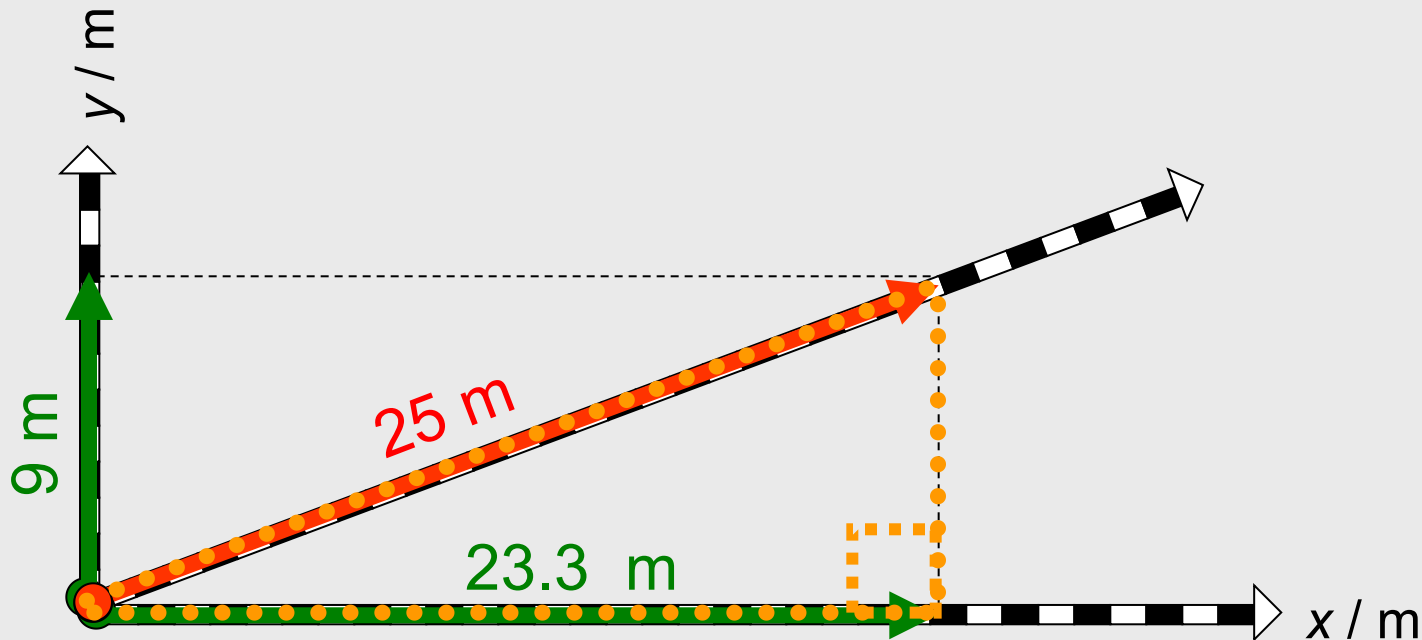
- The green balls are just the shadow of the red ball on each axis. Watch the animation repeatedly and observe how the shadows also have velocities.



3-4 Adding Vectors by Components

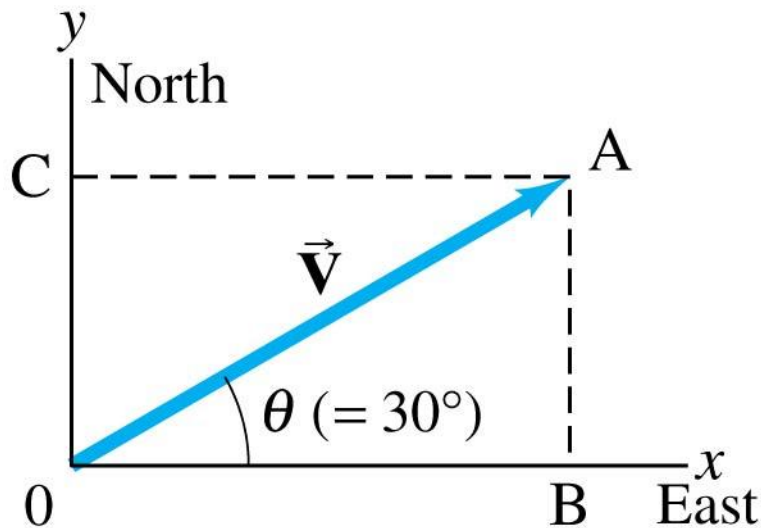
Combination and resolution of vectors

- We can measure each side directly on our scale:
- Note that if we move the 9 m side to the right we complete a **right triangle**.
- Clearly, vectors at an angle can be broken down into the pieces represented by their shadows.

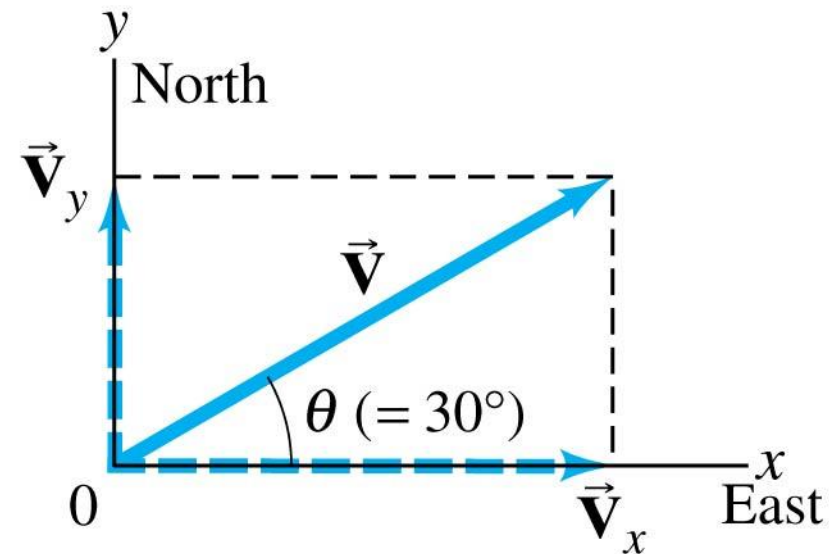


3-4 Adding Vectors by Components

- Any vector can be expressed as the sum of two other vectors, which are called its components. The components are usually chosen to be along two perpendicular directions, such as the x and y axes.
- The process of finding the components is known as **resolving the vector into its components**. Usually, we show vector components as arrows, like vectors, but **dashed**.



(a)

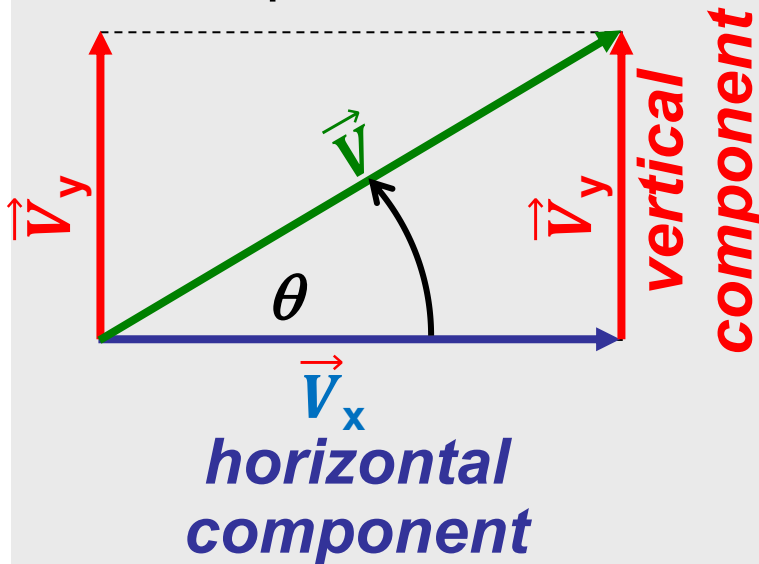


(b)

3-4 Adding Vectors by Components

Combination and resolution of vectors

- Consider a generalized vector $\vec{V}$ as shown below.
- We can break the vector $\vec{V}$ down into its horizontal or x-component $\vec{V}_x$ and its vertical or y-component $\vec{V}_y$.
- We can also sketch in an angle, and perhaps measure it with a protractor.



- In physics and most sciences, we use the Greek letter θ (theta) to represent an angle.

- From Pythagoras we have

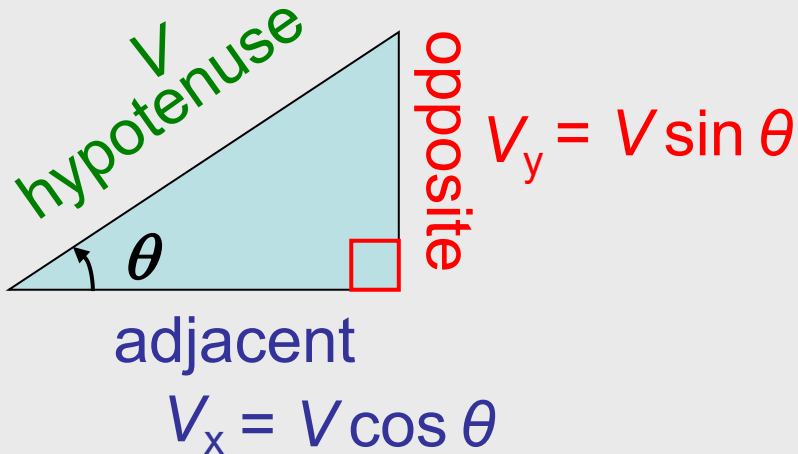
$$V^2 = V_y^2 + V_x^2.$$

3-4 Adding Vectors by Components

Combination and resolution of vectors

- Recall the trigonometry of a right triangle:

$$\sin \theta = \frac{\text{opp}}{\text{hyp}} \quad \frac{V_y}{V} \quad \cos \theta = \frac{\text{adj}}{\text{hyp}} \quad \frac{V_x}{V} \quad \tan \theta = \frac{\text{opp}}{\text{adj}} \quad \frac{V_y}{V_x}$$



s-o-h-c-a-h-t-o-a

trigonometric
ratios

CAUTION! The equations are valid only if θ is the angle $\vec{V}$ makes with the positive x axis.

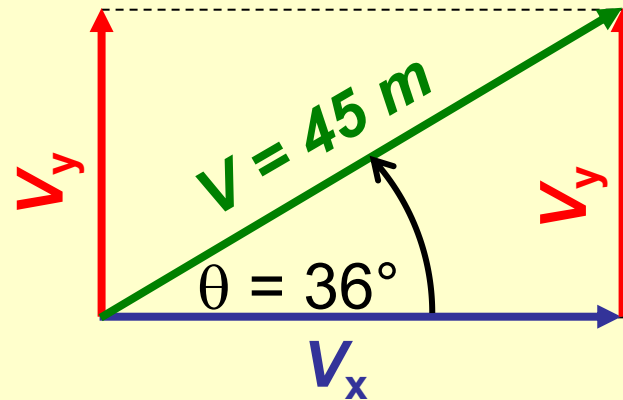
3-4 Adding Vectors by Components

Combination and resolution of vectors

EXAMPLE: A student walks 45 m on a staircase that rises at a 36° angle with respect to the horizontal (the x-axis). Find the x- and y-components of his journey.

SOLUTION: A picture helps.

- $V_x = V \cos \theta$
 $= 45 \cos 36^\circ = 36 \text{ m}$
- $V_y = V \sin \theta$
 $= 45 \sin 36^\circ = 26 \text{ m}$



FYI

- To **resolve** a vector means to break it down into its x- and y-components.

3-4 Adding Vectors by Components

Adding Vectors

We can now discuss how to add vectors using components. The first step is to resolve each vector into its components. Next we can see, using Fig. 3-14, that the addition of any two vectors $\vec{V}_1$ and $\vec{V}_2$ to give a resultant, $\vec{V}_R = \vec{V}_1 + \vec{V}_2$, implies that

$$\begin{aligned} V_{Rx} &= V_{1x} + V_{2x} \\ V_{Ry} &= V_{1y} + V_{2y}. \end{aligned} \quad (3-5)$$

If the magnitude and direction of the resultant vector are desired, they can be obtained using:

$$\begin{aligned} V &= \sqrt{V_x^2 + V_y^2} \\ \tan \theta &= \frac{V_y}{V_x} \end{aligned}$$

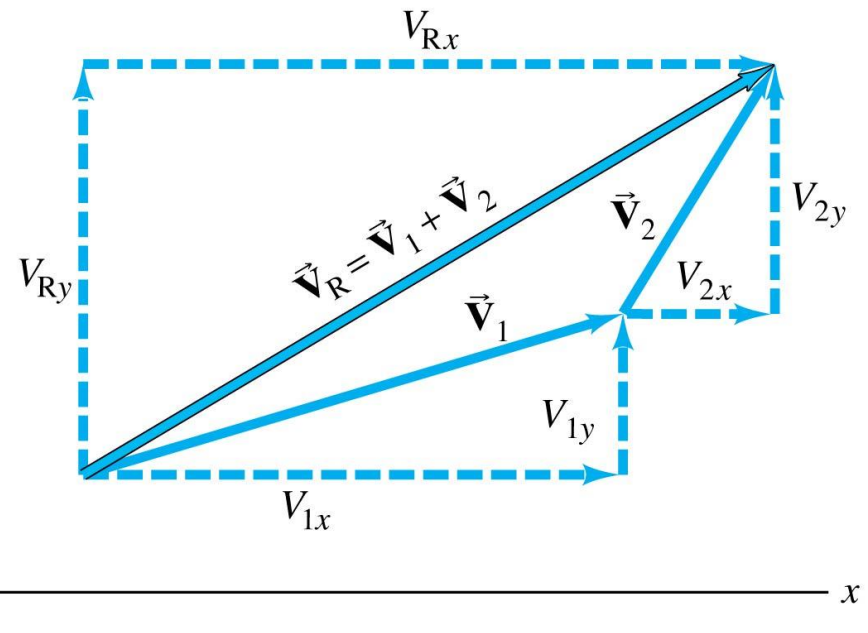


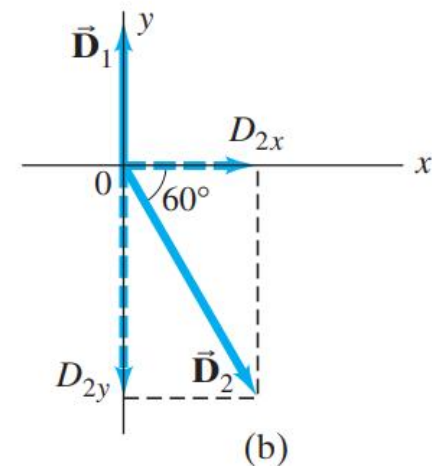
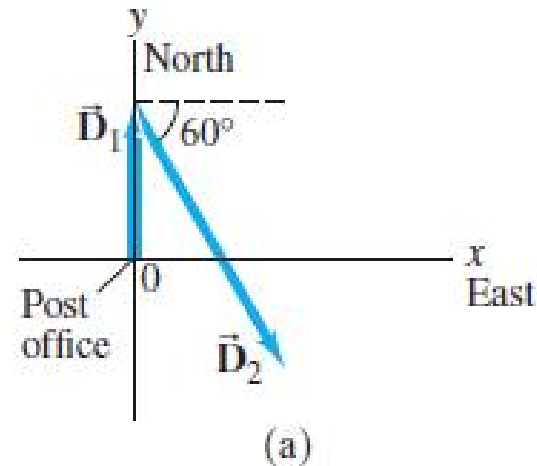
Fig. 3-14

3-4 Adding Vectors by Components

Mail carrier's displacement. A rural mail carrier leaves the post office and drives 22.0 km in a northerly direction. She then drives in a direction 60.0° south of east for 47.0 km. What is her displacement from the post office?

Approach:

1. We choose the positive x axis to be east and the positive y axis to be north, since those are the compass directions used on most maps.
2. The origin of the xy coordinate system is at the post office.
3. We resolve each vector into its x and y components.
4. We add the x components together, and then the y components together, giving us the x and y components of the resultant.



3-4 Adding Vectors by Components

Mail carrier's displacement. A rural mail carrier leaves the post office and drives 22.0 km in a northerly direction. She then drives in a direction 60.0° south of east for 47.0 km. What is her displacement from the post office?

SOLUTION

Fig. 3-15

Since $\mathbf{D}_1$ has magnitude 22.0 km and points north, it has only a y-component

$$D_{1x} = 0, \quad D_{1y} = 22.0 \text{ km.}$$

$\mathbf{D}_2$ has both x and y components:

$$D_{2x} = +(47.0 \text{ km})(\cos 60^\circ) = +(47.0 \text{ km})(0.500) = +23.5 \text{ km}$$

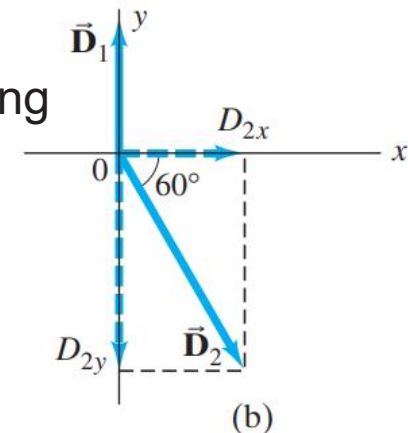
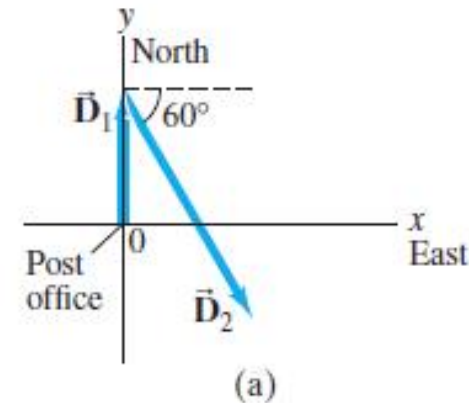
$$D_{2y} = -(47.0 \text{ km})(\sin 60^\circ) = -(47.0 \text{ km})(0.866) = -40.7 \text{ km.}$$

Notice that D_{2y} is negative because this vector component points along the negative y axis.

Then, the resultant vector, has components:

$$D_{Rx} = D_{1x} + D_{2x} = 0 \text{ km} + 23.5 \text{ km} = +23.5 \text{ km}$$

$$D_{Ry} = D_{1y} + D_{2y} = 22.0 \text{ km} + (-40.7 \text{ km}) = -18.7 \text{ km.}$$



3-4 Adding Vectors by Components

Mail carrier's displacement. A rural mail carrier leaves the post office and drives 22.0 km in a northerly direction. She then drives in a direction 60.0° south of east for 47.0 km. What is her displacement from the post office?

SOLUTION (cont..)

This specifies the resultant vector completely:

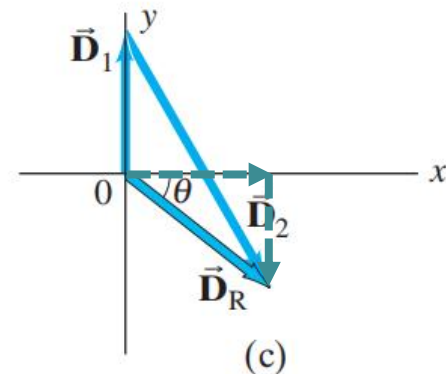
$$D_{Rx} = 23.5 \text{ km}, \quad D_{Ry} = -18.7 \text{ km}.$$

$$D_R = \sqrt{D_{Rx}^2 + D_{Ry}^2} = \sqrt{(23.5 \text{ km})^2 + (-18.7 \text{ km})^2} = 30.0 \text{ km}$$

$$\tan \theta = \frac{D_{Ry}}{D_{Rx}} = \frac{-18.7 \text{ km}}{23.5 \text{ km}} = -0.796.$$

A calculator with a key labeled INV TAN, or ARC TAN, or $\tan^{-1}$ gives $\theta = \tan^{-1}(-0.796) = -38.5^\circ$. The negative sign means $\theta = 38.5^\circ$ below the x axis, Fig. 3-15c. So, the resultant displacement is 30.0 km directed at 38.5° in a southeasterly direction.

Fig. 3-15c



NOTE Always be attentive about the quadrant in which the resultant vector lies. An electronic calculator does not fully give this information, but a good diagram does.

S	A
$\cos \theta = \text{negative}$ $\sin \theta = \text{positive}$ $\tan \theta = \text{negative}$	$\cos \theta = \text{positive}$ $\sin \theta = \text{positive}$ $\tan \theta = \text{positive}$
T	C
$\cos \theta = \text{negative}$ $\sin \theta = \text{negative}$ $\tan \theta = \text{positive}$	$\cos \theta = \text{positive}$ $\sin \theta = \text{negative}$ $\tan \theta = \text{negative}$

3-4 Adding Vectors by Components

Adding vectors:

1. Draw a diagram; add the vectors graphically.
2. Choose x and y axes.
3. Resolve each vector into x and y components.
4. Calculate each component using sines and cosines.
5. Add the components in each direction.
6. To find the length and direction of the vector, use:

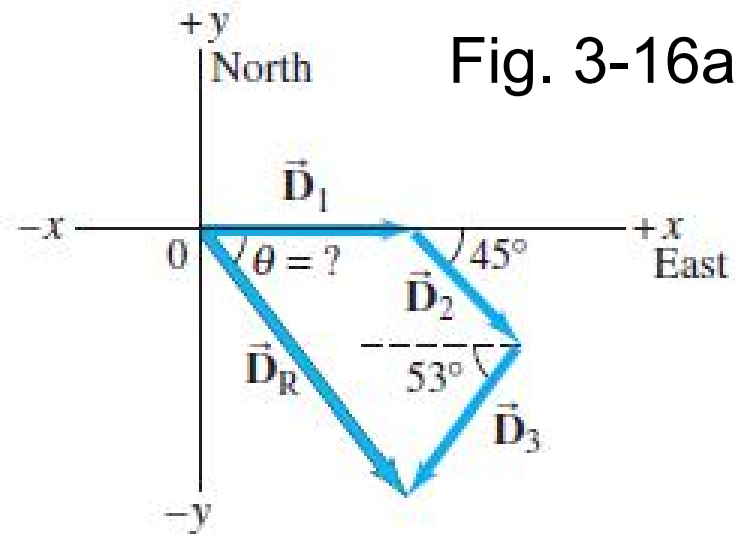
$$V = \sqrt{V_x^2 + V_y^2}$$

$$\tan \theta = \frac{V_y}{V_x}$$

3-4 Adding Vectors by Components

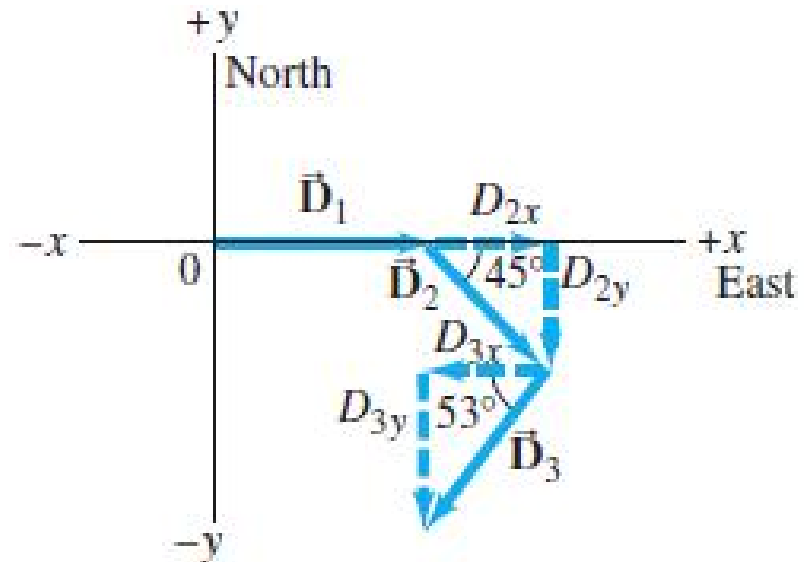
EXAMPLE 3-3 Three short trips. An airplane trip involves three legs, with two stopovers, as shown in Fig. 3-16a. The first leg is due east for 620 km; the second leg is southeast (45°) for 440 km; and the third leg is at 53° south of west, for 550 km, as shown. What is the plane's total displacement?

1. Draw a diagram



2. Choose axes: x is east, y north.

3. Resolve components



Instead of drawing all the vectors starting from a common origin, here we draw the tail-to-tip style.

3-4 Adding Vectors by Components

4. Calculate the components:

$$\vec{D}_1: D_{1x} = +D_1 \cos 0^\circ = D_1 = 620 \text{ km}$$

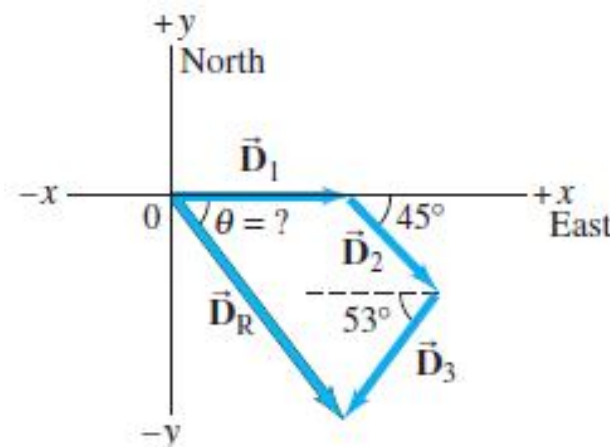
$$D_{1y} = +D_1 \sin 0^\circ = 0 \text{ km}$$

$$\vec{D}_2: D_{2x} = +D_2 \cos 45^\circ = +(440 \text{ km})(0.707) = +311 \text{ km}$$

$$D_{2y} = -D_2 \sin 45^\circ = -(440 \text{ km})(0.707) = -311 \text{ km}$$

$$\vec{D}_3: D_{3x} = -D_3 \cos 53^\circ = -(550 \text{ km})(0.602) = -331 \text{ km}$$

$$D_{3y} = -D_3 \sin 53^\circ = -(550 \text{ km})(0.799) = -439 \text{ km}.$$



5. Add the components

$$D_{Rx} = D_{1x} + D_{2x} + D_{3x} = 620 \text{ km} + 311 \text{ km} - 331 \text{ km} = 600 \text{ km}$$

$$D_{Ry} = D_{1y} + D_{2y} + D_{3y} = 0 \text{ km} - 311 \text{ km} - 439 \text{ km} = -750 \text{ km}.$$

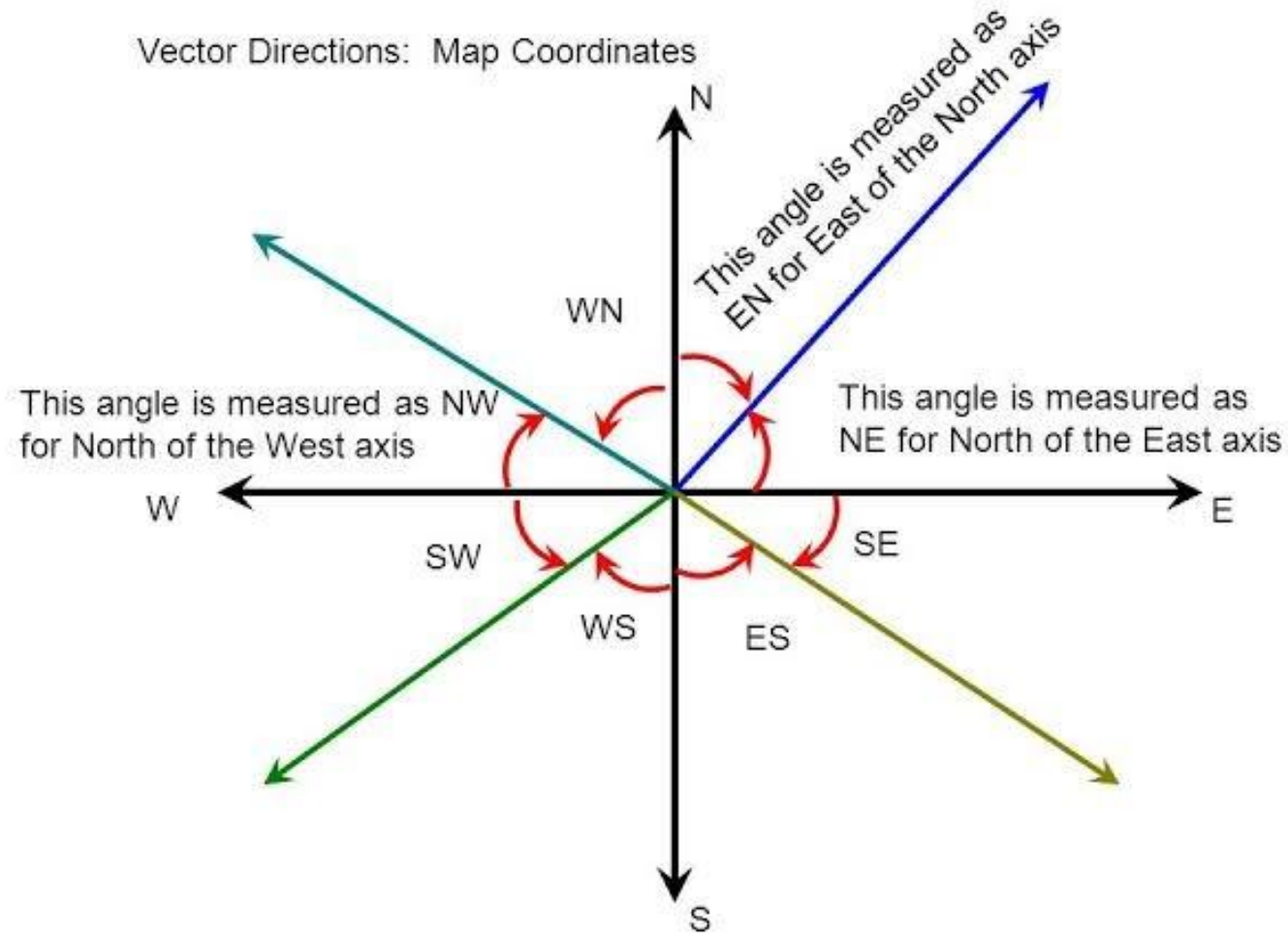
6. Magnitude and direction

$$D_R = \sqrt{D_{Rx}^2 + D_{Ry}^2} = \sqrt{(600)^2 + (-750)^2} \text{ km} = 960 \text{ km}$$

$$\tan \theta = \frac{D_{Ry}}{D_{Rx}} = \frac{-750 \text{ km}}{600 \text{ km}} = -1.25, \quad \text{so } \theta = -51^\circ.$$

Thus, the total displacement has magnitude 960 km and points 51° below the x axis (south of east), as was shown in our original sketch,

3-4 Adding Vectors by Components



3-5 Projectile Motion

Projectile motion

- A **projectile** is an object that has been given an initial velocity by some sort of short-lived force, and then moves through the air under the influence of gravity.
- Baseballs, stones, or bullets are all examples of projectiles executing **projectile motion**.
- You know that all objects moving through air feel an air resistance (recall sticking your hand out of the window of a moving car).



FYI

- We will ignore air resistance in the discussion that follows...

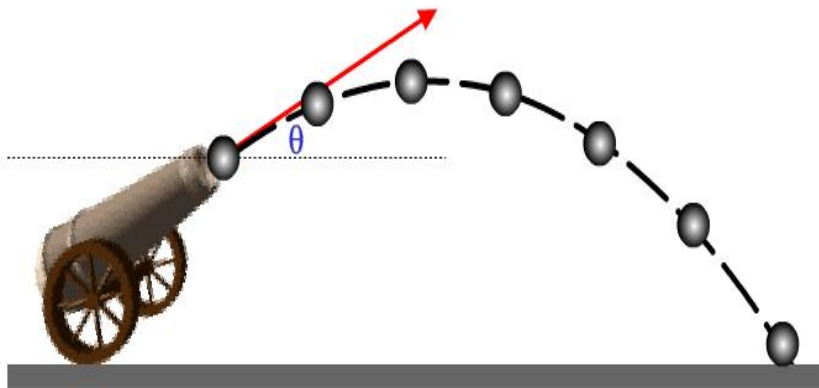
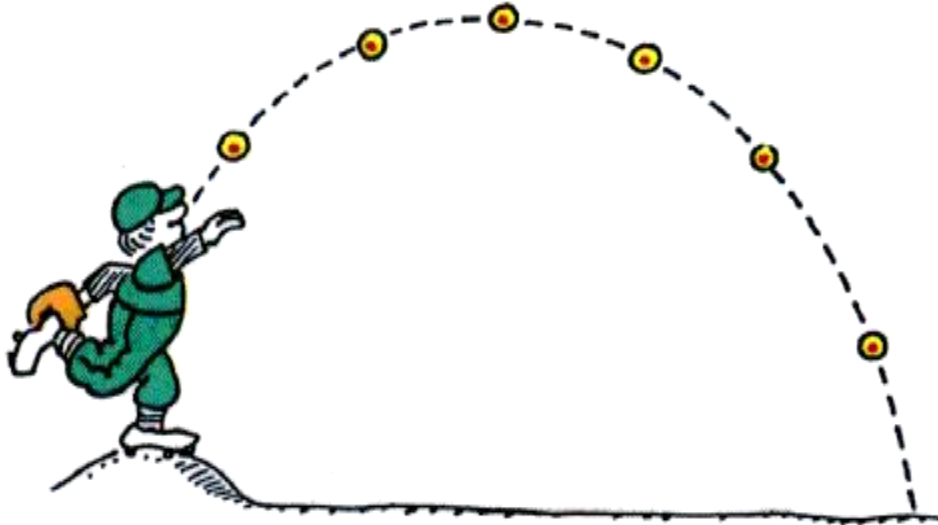


3-5 Projectile Motion

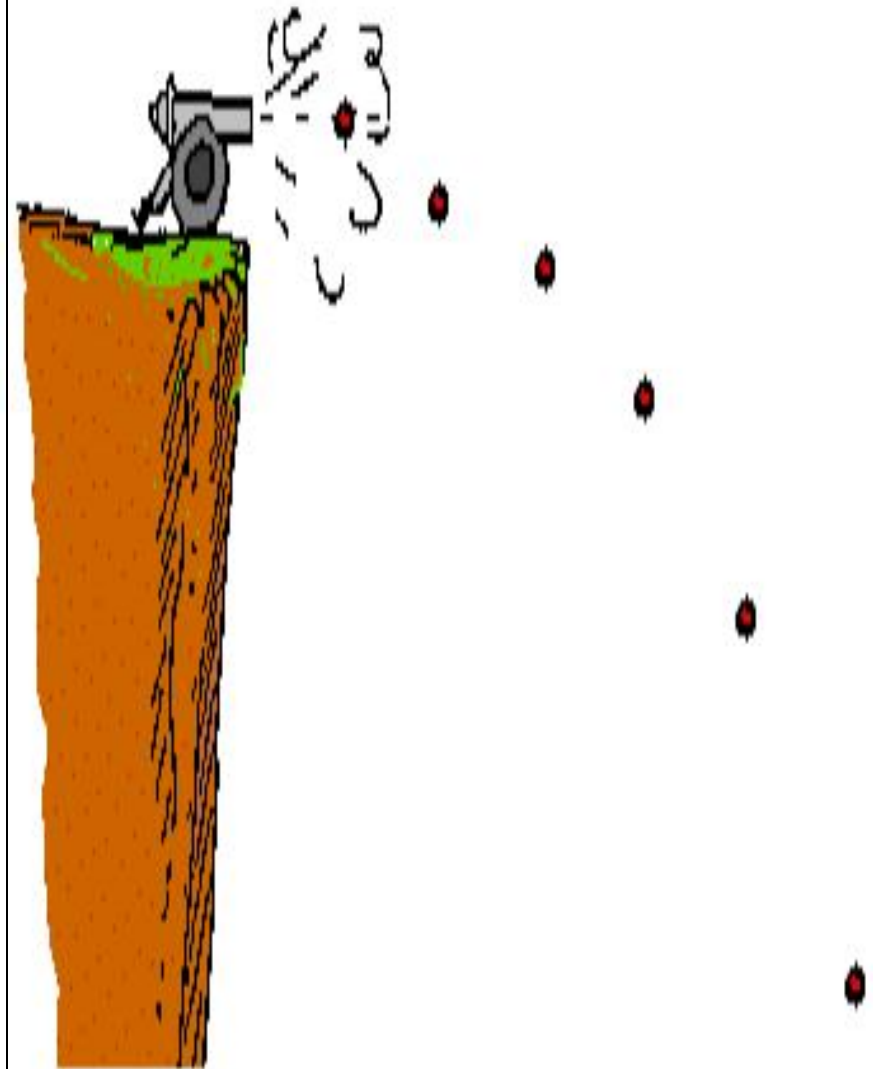
- 2-dimensional motion.
- consists of **horizontal and vertical components of motion**. These 2 components of motion MUST be discussed separately.
- Assumptions of projectile motion:
 - 1) free fall acceleration, g is constant and is always directed downward.
 - 2) Neglect air resistance.



(1) Projection at an angle



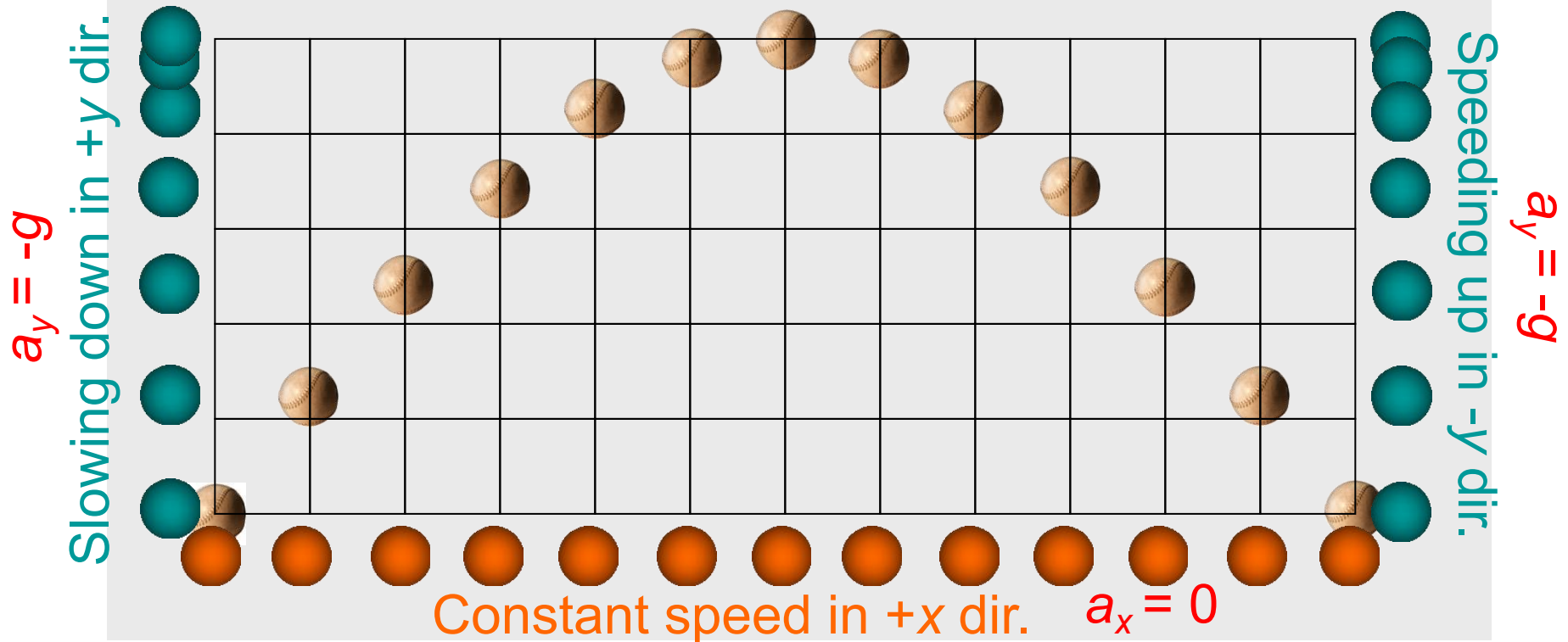
(2) Horizontal Projection



3-5 Projectile Motion-Projection at Angle

Analysing projectile motion

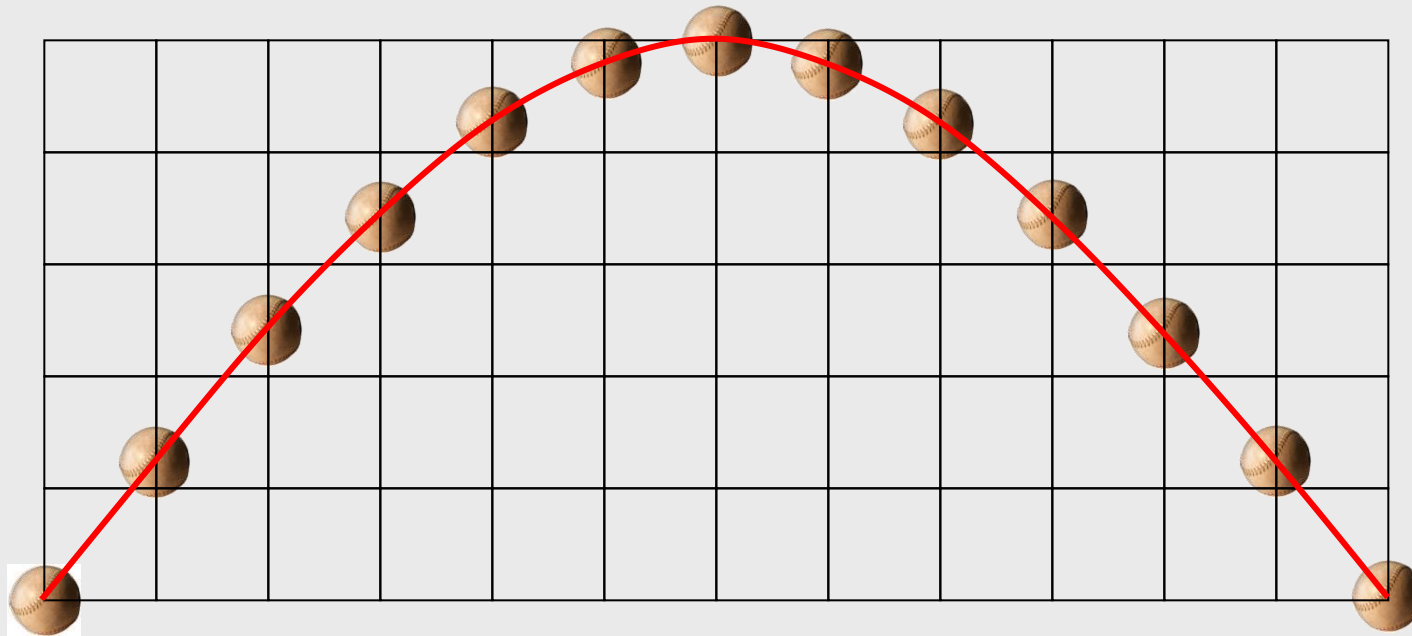
- Regardless of the air resistance, the vertical and the horizontal components of velocity of an object in projectile motion are independent.



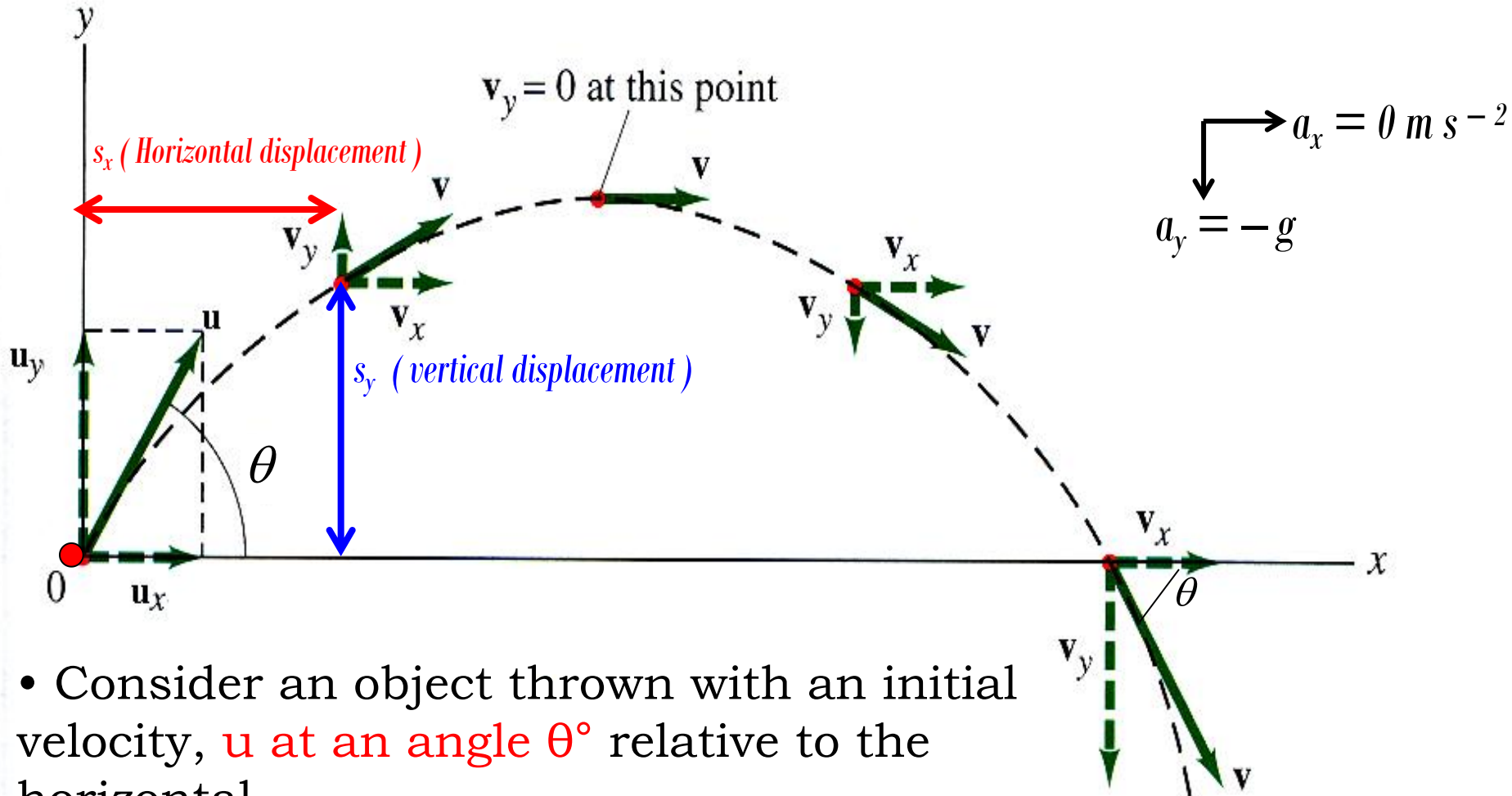
3-5 Projectile Motion-Projection at Angle

Analysing projectile motion

- The trajectory of a projectile in the absence of air is **parabolic**. Know this!



3-5 Projectile Motion-Projection at Angle



- Consider an object thrown with an initial velocity, u at an angle θ° relative to the horizontal.
- The object move upward or downward it also moving horizontally.

-- the path of motion : **parabolic arc**



Horizontal & vertical motions are independent and discussed separately in calculation.

-- The **initial velocity, u** is resolved into horizontal and vertical components :

$$u_x = u \cos \theta$$

$$u_y = u \sin \theta$$

-- In vertical (y axis), **gravity acts downwards** ($a_y = -g$) $\rightarrow$ object does **free fall** motion.

-- **y -component** of the **velocity changes with time**.

$$\Delta y = u_y t - \frac{1}{2} g t^2$$

$$v_y = u_y - g t$$

$$v_y^2 = u_y^2 - 2 g s_y$$

*Kinematic equations
for constant
acceleration are
applied!!*

- The vertical displacement, s_y (when $s_o=0$) is given by:

$$s_y - s_o = u_y t - \frac{1}{2} g t^2$$

$$s_y = u_y t - \frac{1}{2} g t^2$$

*s_y also known
as height*

- Gravitational force does not act horizontally -- **no acceleration in horizontal** direction ($a_x = 0$), object travels in horizontal direction with uniform velocity.
- Horizontal component of the velocity, v_x is constant.

From: $v_x = u_x + \cancel{a_x t}$

$$v_x = u_x$$

- With $a_x = 0$ and $s_o = 0$, the horizontal displacement, s_x is given by:

$$s_x = u_x t + \frac{1}{2} \cancel{a_x} t^2$$

$$\boxed{s_x = u_x t}$$

- The magnitude of the instantaneous velocity v at any point is given by :

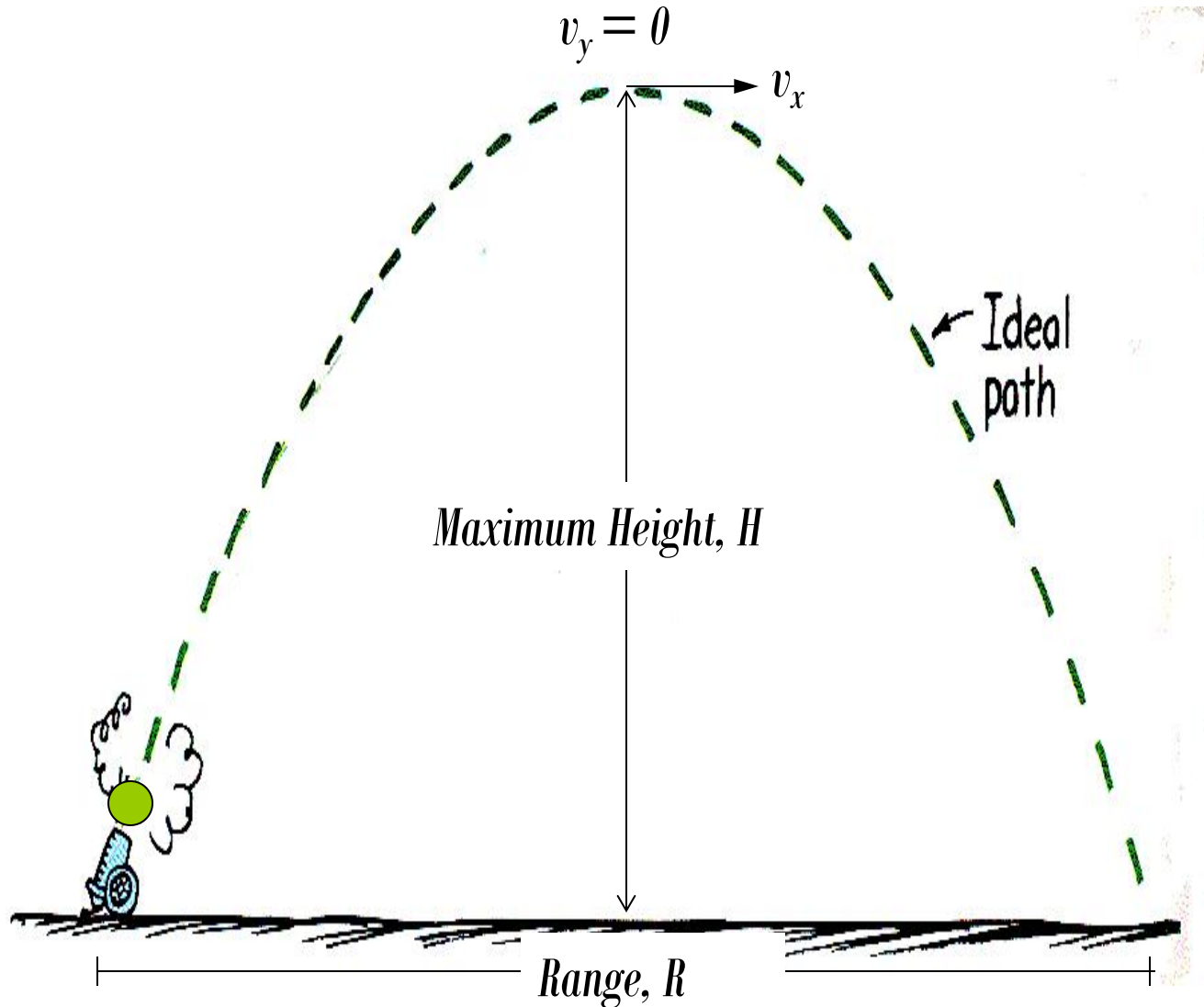
$$\boxed{v = \sqrt{v_x^2 + v_y^2}}$$

Direction of the velocity :

$$\boxed{\tan \theta = \frac{v_y}{v_x} \Rightarrow \theta \text{ is from horizontal line}}$$



For projectile returns to the same vertical level at which it was launched



- At the maximum height (H), the velocity (y-comp) $v_y = 0$.

Using : $v_y^2 = u_y^2 - 2g s_y$

$$0 = (u \sin \theta)^2 - 2gH$$

$$\boxed{H = \frac{u^2 \sin^2 \theta}{2g}} \quad [\text{Maximum Height}]$$

- If t_H = Time taken by the object to reach the maximum height, H

Using :

$$v_y = u_y - g t$$

$$0 = (u \sin \theta) - g t_H$$

$$\boxed{t_H = \frac{u \sin \theta}{g}} \quad [\text{time to reach maximum height}]$$

Let T = time of flight

When the body lands on the ground, $\Delta y = 0$

Using :

$$\Delta y = u_y - \frac{1}{2}gt^2$$

$$0 = (u \sin \theta)T - \frac{1}{2}gT^2$$

$$\boxed{T = \frac{2u \sin \theta}{g}}$$

$$T = 2\left(\frac{u \sin \theta}{g}\right) \Rightarrow \boxed{T = 2t_H}$$

* As the path is symmetrical, time in **going up** is equal to the time in **coming down**.

3-5 –Projectile Motion – Horizontal Range

Level Horizontal Range

The total distance travelled in horizontal is called the horizontal **range**, R .

Derivation formula [$y = y_0$]

$$\Delta y = u_y t - \frac{1}{2} g t^2$$

$$\Delta y = 0$$

$$u_y t = \frac{1}{2} g t^2$$

$$u_y = \frac{1}{2} g t$$

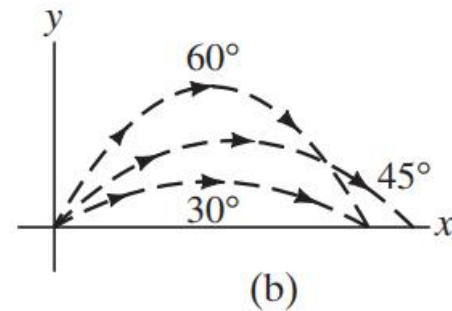
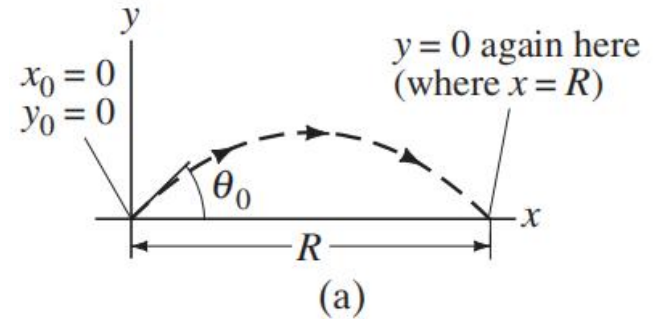
$$t = \frac{2u_y}{g}$$

$$\Delta x = u_x t$$

$$R = u_x t$$

$$R = u_x \left(\frac{2u_y}{g} \right)$$
$$= \frac{2u_x u_y}{g}$$

$$= \frac{2u^2 \sin \theta \cos \theta}{g}$$
$$= \frac{u^2 \sin 2\theta}{g}$$



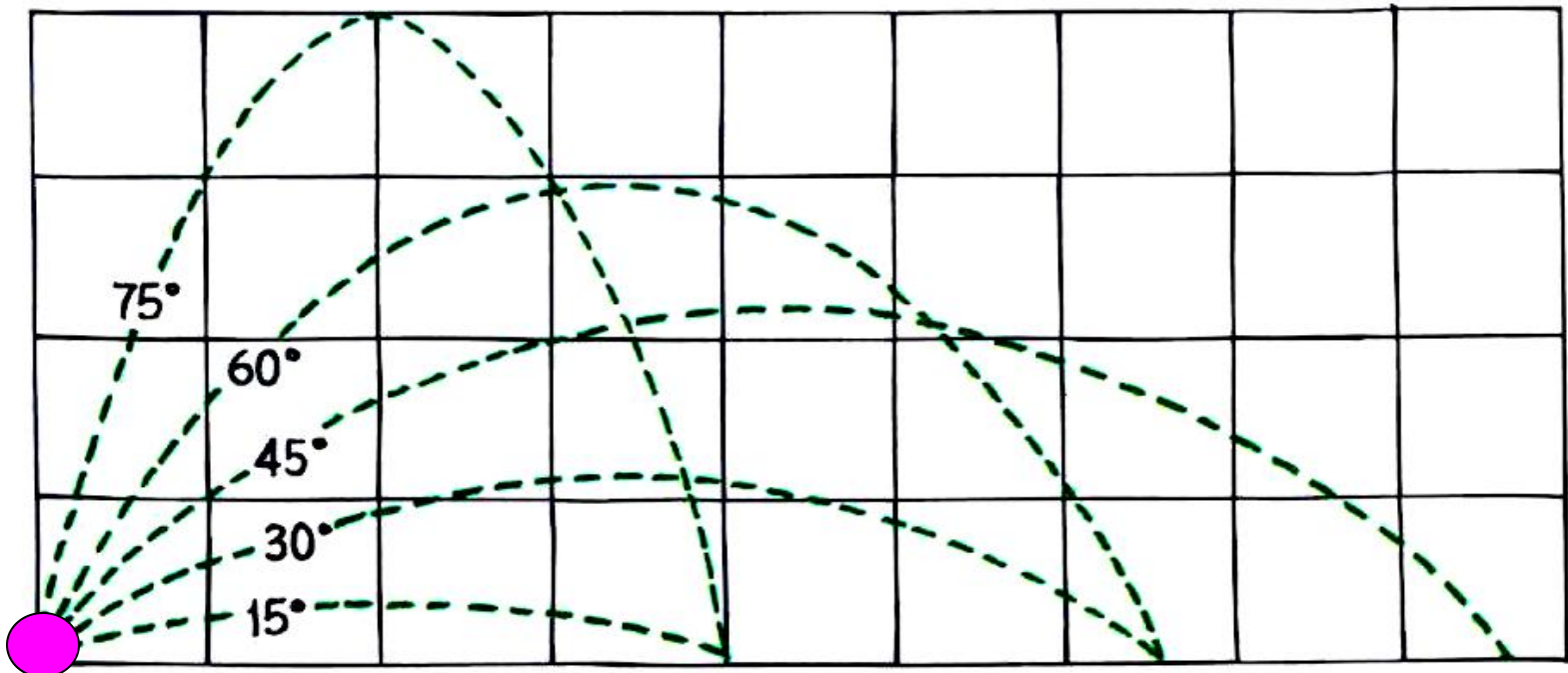
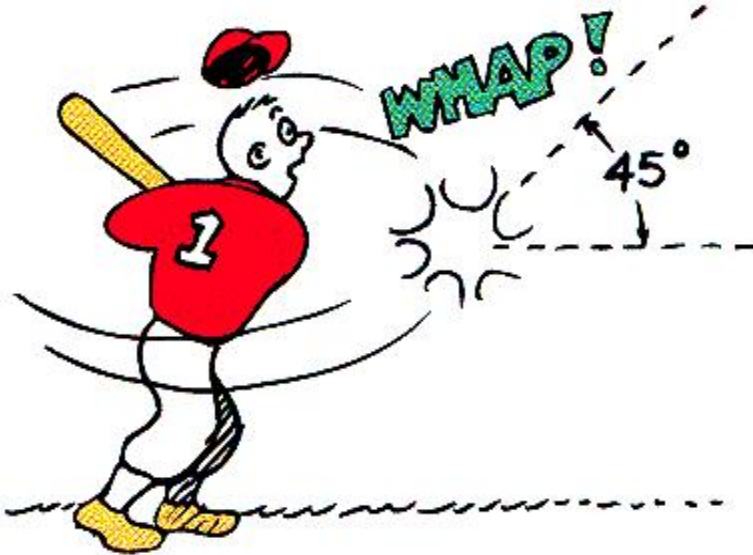
$$\text{where } (2 \sin \theta \cos \theta = \sin 2\theta)$$

Maximum range, $R_{\max}$ at a particular speed is obtained when :

$$\sin 2\theta = 1$$

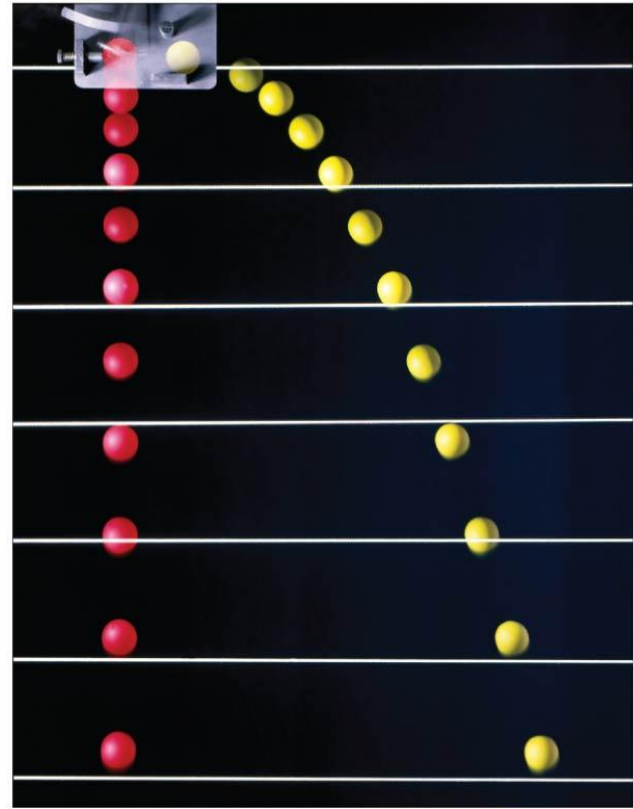
$$2\theta = 90^\circ$$

$$\Rightarrow \theta = 45^\circ$$

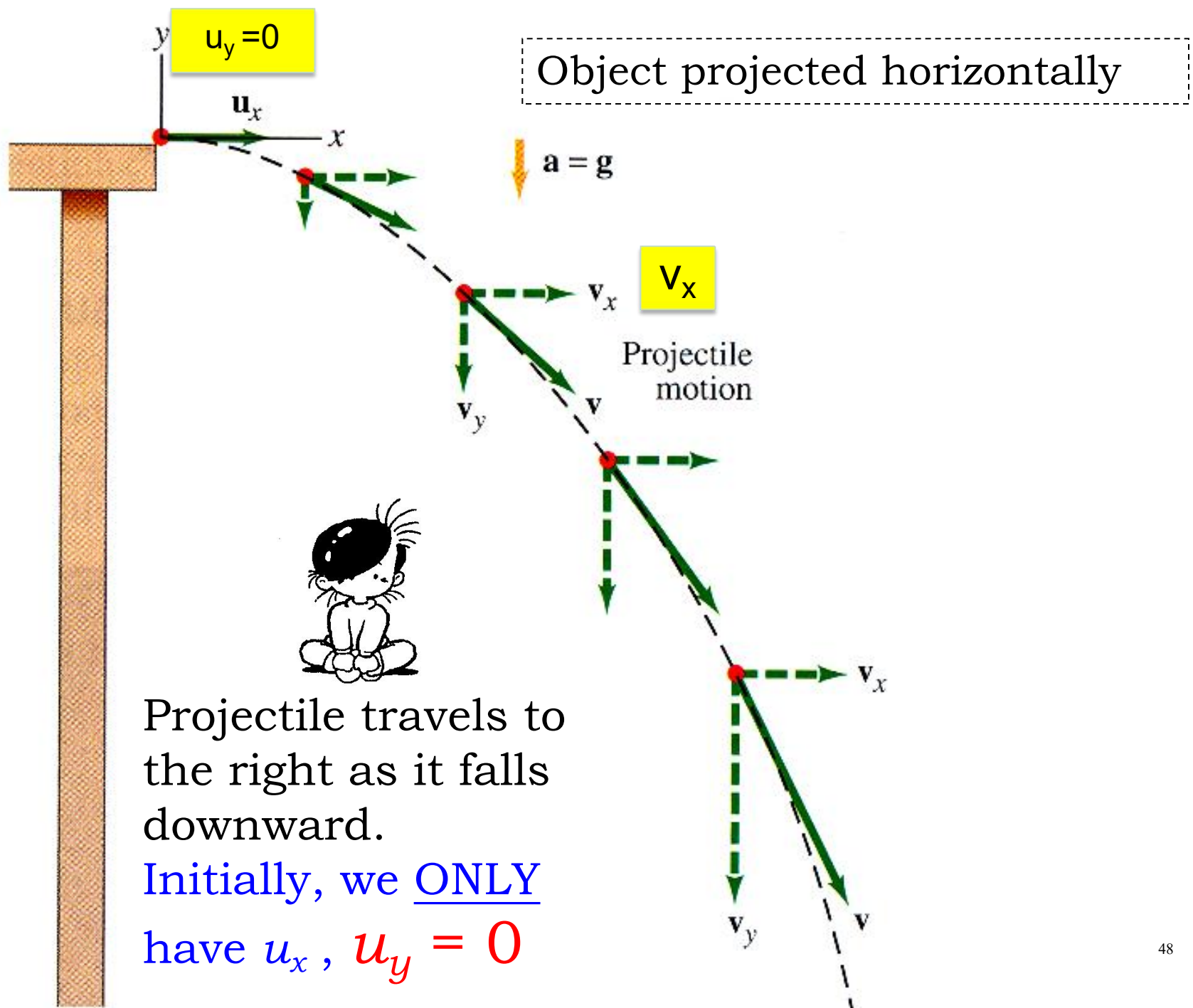


3-5 Projectile Motion– Projected horizontally

- An object projected horizontally will reach the ground in the same time as an object dropped vertically. This is because the vertical motions are the same in both cases*



Multiple-exposure photograph showing positions of two balls at equal time intervals. One ball was dropped from rest at the same time the other ball was projected horizontally outward. The vertical position of each ball is seen to be the same at each instant.



3-6 Solving Projectile Motion Problems

Analyzing projectile motion

- Recall the kinematic equations:

$$s = ut + (1/2)at^2 \quad \text{Displacement}$$

$$v = u + at \quad \text{Velocity}$$

$$v^2 = u^2 + 2as$$

kinematic
equations 1D

a is constant

- Since we worked only in 1D at the time, we didn't have to distinguish between x and y in these equations.
- Now we appropriately modify the above to meet our new requirements of simultaneous equations:

$$s_x = u_x t + (1/2)a_x t^2$$

$$v_x = u_x + a_x t$$

$$s_y = u_y t + (1/2)a_y t^2$$

$$v_y = u_y + a_y t$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

kinematic
equations 2D

a_x and a_y are
constant

3-6 Solving Projectile Motion Problems

Analysing projectile motion

$$\Delta x @ s_x = u_x t + (1/2) a_x^0 t^2$$

$$v_x = u_x + a_x^0 t$$

$$\Delta y @ s_y = u_y t + (1/2) a_y t^2$$

$$v_y = u_y + a_y t$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

kinematic
equations 2D

a_x and a_y are
constant

PRACTICE: Show that the reduced equations for projectile motion are

$$\Delta x = u_x t$$

$$\Delta y = u_y t - 4.9 t^2$$

$$v_x = u_x$$

$$v_y = u_y - 9.80 t$$

$$v_y^2 = u_y^2 - 2(9.80) \Delta y$$

reduced equations of
projectile motion

SOLUTION:

- $a_x = 0$ in the absence of air resistance.
- $a_y = -9.80 \text{ m/s}^2$ in the absence of air resistance.

3-6 Solving Projectile Motion Problems

Analysing projectile motion

$$\Delta x = u_x t \quad \Delta y = u_y t - 4.9 t^2$$

$$v_x = u_x \quad v_y = u_y - 9.80 t$$

reduced equations of
projectile motion

EXAMPLE: Use the reduced equations above to prove that projectile motion is parabolic in nature.

SOLUTION: Just solve for t in the first equation and substitute it into the second equation.

$$\Delta x = u_x t \text{ becomes } t = x / u_x \text{ so that } t^2 = x^2 / u_x^2.$$

Then since $y = u_y t - 4.9 t^2$, we have

$$y = (u_y / u_x) x - (4.9 / u_x^2) x^2.$$

FYI

- The general equation of a parabola is $y = Ax + Bx^2$.
- In this case, $A = u_y / u_x$ and $B = -4.9 / u_x^2$.

Equations in Projectile Motion

Horizontal (x motion)

$$u_x = u \cos \theta$$

$$a_x = 0$$

$$v_x = u_x$$

$$s_x = u_x t$$



Vertical (y motion)

$$u_y = u \sin \theta$$

$$a_y = -g$$

$$v_y = u_y - g t$$

$$v_y^2 = u_y^2 - 2 g s_y$$

$$s_y = u_y t - \frac{1}{2} g t^2$$

3-6 – Solving Projectile Motion Problems

Solving Problems involving projectile motion can require creativity, and cannot be done just by following some rules. Certainly you must avoid just plugging numbers into equations that seem to “work.”

1. Read carefully; choose the object (or objects).
2. Draw a careful diagram.
3. Choose an origin and an xy coordinate system.
4. Decide on the time interval. The x and y motions are connected by the common time.
5. Examine the horizontal (x) and vertical (y) motions separately.
6. List the known and unknown quantities
7. Apply the relevant equations, combining equations if necessary.
8. You may need to combine components of a vector to get magnitude and direction.

3-6 – Solving Projectile Motion Problems

Analysing projectile motion

$$\Delta x = u_x t \quad \Delta y = u_y t - 4.9 t^2$$

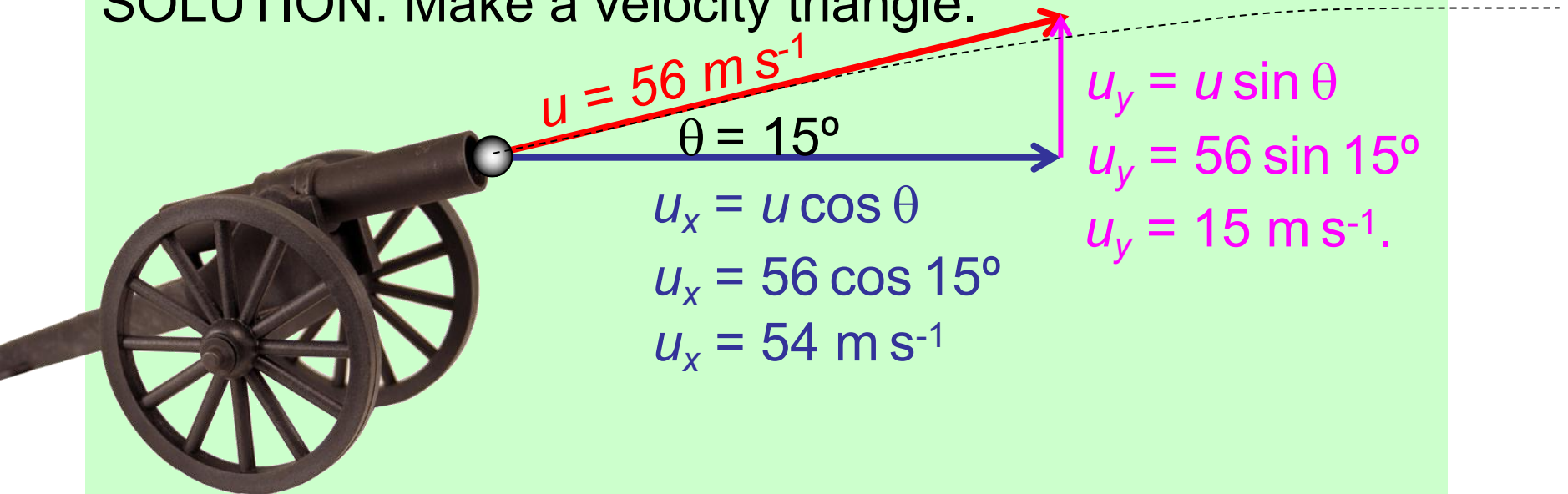
$$v_x = u_x \quad v_y = u_y - 9.80 t$$

reduced equations of
projectile motion

PRACTICE: A cannon fires a projectile with a muzzle velocity of 56 ms^{-1} at an angle of inclination of 15° .

(a) What are u_x and u_y ?

SOLUTION: Make a velocity triangle.



3-6 – Solving Projectile Motion Problems



Analysing projectile motion

$$\Delta x = u_x t \quad \Delta y = u_y t - 4.9t^2$$

$$v_x = u_x \quad v_y = u_y - 9.8t$$

reduced equations of
projectile motion

PRACTICE: A cannon fires a projectile with a muzzle velocity of 56 ms^{-1} at an angle of inclination of 15° .

(b) What are the tailored equations of motion?

(c) When will the ball reach its maximum height?

SOLUTION: (b) Just substitute $u_x = 54$ and $u_y = 15$:

$$\Delta x = 54t \quad \Delta y = 15t - 4.9t^2$$

$$v_x = 54 \quad v_y = 15 - 9.8t$$

tailored equations for
this particular projectile

(c) At the maximum height, $v_y = 0$. Why? Thus

$v_y = 15 - 9.8t$ becomes $0 = 15 - 9.8t$ so that

$$9.8t = 15 \text{ or } t = 1.53 \text{ s.}$$

3-6 – Solving Projectile Motion Problems



Analysing projectile motion

$$\begin{array}{ll} \Delta x = 54t & \Delta y = 15t - 4.9t^2 \\ v_x = 54 & v_y = 15 - 9.80t \end{array}$$

tailored equations for
this particular projectile

PRACTICE: A cannon fires a projectile with a muzzle velocity of 56 ms^{-1} at an angle of inclination of 15° .

(d) How far from the muzzle will the ball be when it reaches the height of the muzzle at the end of its trajectory?

SOLUTION:

From symmetry $t_{up} = t_{down} = 1.53 \text{ s}$ so $t = 3.06 \text{ s}$.

Thus

$$\Delta x = 54t$$

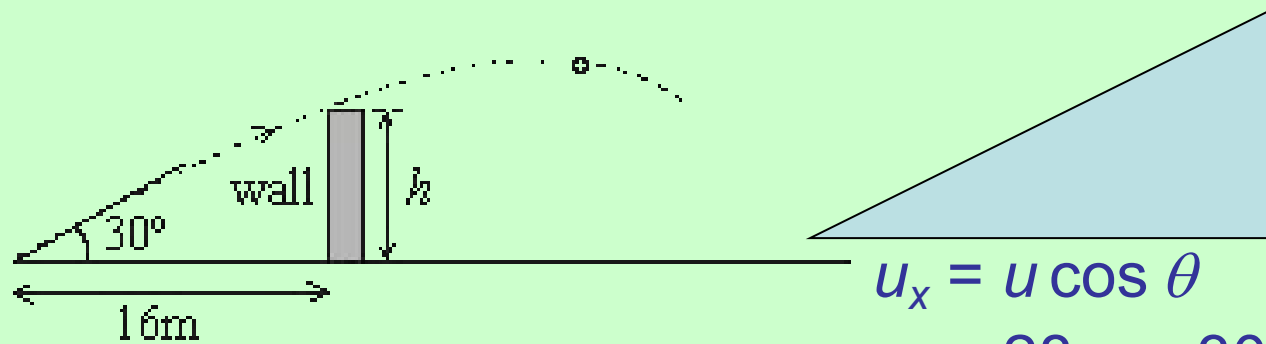
$$\Delta x = 54(3.06)$$

$$\Delta x = 165.2 \text{ m.}$$

3-6 – Solving Projectile Motion Problems

Analysing projectile motion

A ball is projected from ground level with a speed of 28 m s^{-1} at an angle of 30° to the horizontal as shown below.



$$u_y = u \sin \theta$$
$$u_y = 28 \sin 30^\circ$$

$$u_x = u \cos \theta$$

$$u_x = 28 \cos 30^\circ$$

There is a wall of height h at a distance of 16 m from the point of projection of the ball. Air resistance is negligible.

- (a) Calculate the initial magnitudes of
- (i) the horizontal velocity of the ball;

$$u_x = 24 \text{ m s}^{-1}.$$

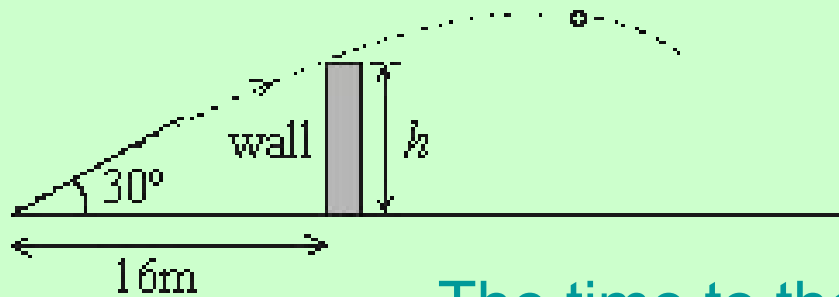
- (ii) the vertical velocity of the ball.

$$u_y = 14 \text{ m s}^{-1}.$$

3-6 – Solving Projectile Motion Problems

Analysing projectile motion

A ball is projected from ground level with a speed of 28 m s^{-1} at an angle of 30° to the horizontal as shown below.



$$\Delta x = u_x t$$

$$16 = 24t$$

$$t = 16 / 24 = 0.67$$

• The time to the wall is found from Δx ...

There is a wall of height h at a distance of 16 m from the point of projection of the ball. Air resistance is negligible.

(b) The ball just passes over the wall. Determine the maximum height of the wall.

$$\Delta y = u_y t - 4.9t^2$$

$$\Delta y = 14t - 4.9t^2$$

$$\Delta y = 14(0.67) - 4.9(0.67)^2 = 7.2 \text{ m.}$$

3-6 – Solving Projectile Motion Problems

Analysing projectile motion

Which **one** of the following is a true statement concerning the vertical component of the velocity and the acceleration of a projectile when it is at its maximum height? (*The acceleration of free fall is g .*)

	Vertical component of velocity	Acceleration
A.	maximum	zero
B.	maximum	g
C.	zero	zero
D.	zero	g

- The acceleration is ALWAYS g for projectile motion—since it is caused by Earth and its field.
- At the maximum height the projectile switches from upward to downward motion. $v_y = 0$ at switch.

3-6 – Solving Projectile Motion Problems

Analysing projectile motion

A stone is thrown at an angle to the horizontal. Ignoring air resistance, the horizontal component of the initial velocity of the stone determines the value of

- A. range only.
- ~~B. maximum height only.~~
- ~~C. range and maximum height.~~
- ~~D. range and time of flight.~~

- The flight time is limited by the y motion.

- The maximum height is limited by the y motion.

3-6 – Solving Projectile Motion Problems

Analysing projectile motion

A stone is projected horizontally from the top of a cliff. Neglecting air resistance, which *one* of the following correctly describes what happens to the horizontal component of velocity and to the vertical component of velocity?

	Horizontal component of velocity	Vertical component of velocity
A.	Decreases	Increases
B.	Decreases	Constant
C.	Constant	Constant
D.	Constant	Increases

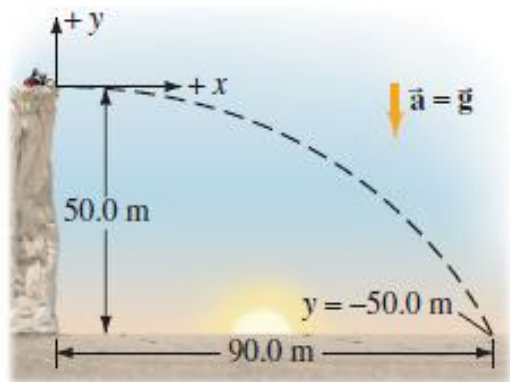
- The horizontal component of velocity is $v_x = u_x$ which is **CONSTANT**.
- The vertical component of velocity is $v_y = u_y - 10t$, which is **INCREASING** (negatively).

3-6 – Solving Projectile Motion Problems_pg 61

Driving off a cliff. A movie stunt driver on a motorcycle speeds horizontally off a 50.0-m-high cliff. How fast must the motorcycle leave the cliff top to land on level ground below, 90.0 m from the base of the cliff where the cameras are? Ignore air resistance.

Approach

- Step 1 and 2: Read, choose the object, and draw a diagram. Our object is the motorcycle and driver, ~~Take choice as isolated in the system.~~
- 3. Choose a coordinate system. ($x_0 = 0, y_0 = 0$)
- 4. Choose a time interval. At $t=0, x_0 = 0, y_0 = 0$. End of time interval just before the motorcycle ~~to examine the ground by location.~~
- 5. Examine the ground by location.
- 6. List knowns and unknowns.
- 7. Apply relevant equations.



Horizontal Motion ($a_x = 0, v_x = \text{constant}$)	Vertical Motion† ($a_y = -g = \text{constant}$)
$v_x = v_{x0}$ $x = x_0 + v_{x0}t$	$v_y = v_{y0} - gt$ $y = y_0 + v_{y0}t - \frac{1}{2}gt^2$ $v_y^2 = v_{y0}^2 - 2g(y - y_0)$

Known	Unknown
$x_0 = y_0 = 0$ $x = 90.0 \text{ m}$ $y = -50.0 \text{ m}$ $a_x = 0$ $a_y = -g = -9.80 \text{ m/s}^2$ $v_{y0} = 0$	v_{x0} t

Initial velocity, v_{x0} ?

$$y = 0 + 0 + \frac{1}{2}(-g)t^2 \quad t = \sqrt{\frac{2y}{-g}} = \sqrt{\frac{2(-50)}{-(9.8)}}$$
$$y = -\frac{1}{2}gt^2 \quad t = 3.19 \text{ s}$$

Remember! The x and y motions are connected by the

Looking for v_{x0}

With known values

$$t = 3.19 \text{ s}$$

$$x_0 = 0$$

$$x = 90 \text{ m}$$

$$a_x = 0$$

Horizontal Motion ($a_x = 0$, $v_x = \text{constant}$)	Vertical Motion [†] ($a_y = -g = \text{constant}$)
$v_x = v_{x0}$ $x = x_0 + v_{x0}t$	$v_y = v_{y0} - gt$ $y = y_0 + v_{y0}t - \frac{1}{2}gt^2$ $v_y^2 = v_{y0}^2 - 2g(y - y_0)$

$$x = x_0 + v_{0x}t$$

$$v_{0x} = \frac{x - x_0}{t}$$

$$v_{0x} = \frac{90 - 0}{3.19}$$

$$v_{0x} = 28.2 \text{ m/s}$$

3-6 – Solving Projectile Motion Problems_pg 62

A kicked football. A kicked football leaves the ground at an angle $\theta_0 = 37.0^\circ$ with a velocity of 20.0 m/s , as shown in Fig. 3–23. Calculate (a) the maximum height, (b) the time of travel before the football hits the ground, and (c) how far away it hits the ground. Assume the ball leaves the foot at ground level, and ignore air resistance and rotation of the ball.

Given:

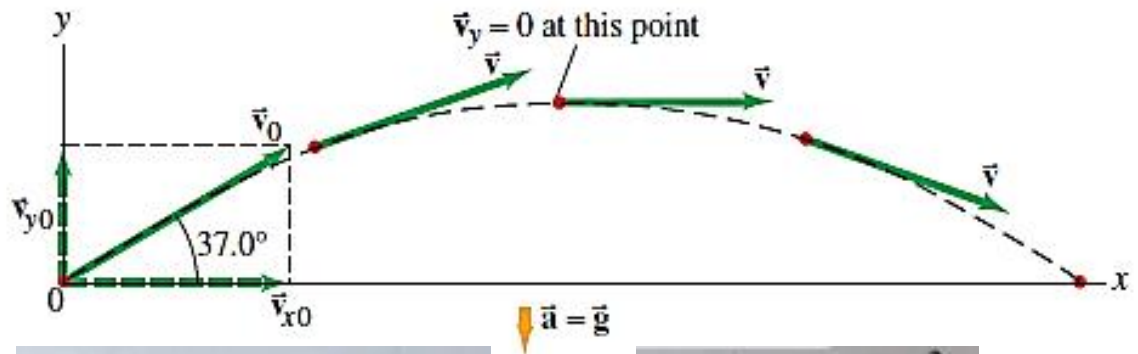
$$\theta = 37^\circ$$

$$u = 20 \text{ ms}^{-1}$$

a) max height?

b) t before ball hit the ground

c) find x .



$$a) v_y^2 = u_y^2 - 2g(\Delta y)$$

$$\Delta y = H.$$

$$v_y^2 = u_y^2 - 2gH.$$

$$2gH = u_y^2$$

$$H = \frac{u_y^2}{2g}$$

$$\begin{aligned} &= \frac{(u \sin \theta)^2}{2g} \\ &= \frac{u^2 \sin^2 \theta}{2g} \\ &= \frac{(12)^2}{2(9.8)} \\ &= 7.35 \text{ m} \end{aligned}$$

b) time of flight?

$$\Delta y = u_y t - \frac{1}{2} g t^2$$

$$y = 0, y_0 = 0 \therefore \Delta y = 0$$

$$u_y t = \frac{1}{2} g t^2$$

$$u_y = \frac{1}{2} g t$$

$$t = \frac{2 u_y}{g}$$

$$= \frac{2(12)}{9.8}$$

$$= 2.45 \text{ s} \# \text{ (total travel time)}$$

c) $x = ?$

$$\Delta x = u_x t$$

$$x_0 = 0, t = 2.45 \text{ s}, u_x = 16 \text{ m s}^{-1}$$

$$\rightarrow x - 0 = 16(2.45)$$

3-6 – Solving Projectile Motion Problems_pg 63

Range of a cannon ball. Suppose one of Napoleon's cannons had a muzzle speed, v_0 , of 60.0 m/s. At what angle should it have been aimed (ignore air resistance) to strike a target 320 m away?

$$R = \frac{u^2 \sin 2\theta}{g}$$

$$320 = \frac{60^2 \sin 2\theta}{9.8}$$

$$\sin 2\theta = \frac{320(9.8)}{60^2}$$

$$\sin 2\theta = 0.871$$

$$\theta = 30.3^\circ \#$$

$$\text{Given: } V_0 = u = 60 \text{ m/s.}$$

$$R = 320 \text{ m.}$$

$$\theta = ?$$

3-6 – Solving Projectile Motion Problems_pg 64

A punt. Suppose the football in Example 3–6 was punted, and left the punter's foot at a height of 1.00 m above the ground. How far did the football travel before hitting the ground? Set $x_0 = 0$, $y_0 = 0$.

Given:

$$\theta = 37^\circ$$

$$u = 20 \text{ m/s}$$

$$y_0 = 0, y = -1 \text{ m (ball hits the ground below its starting point of } y_0 = 0)$$

Should use $x = u_x t$

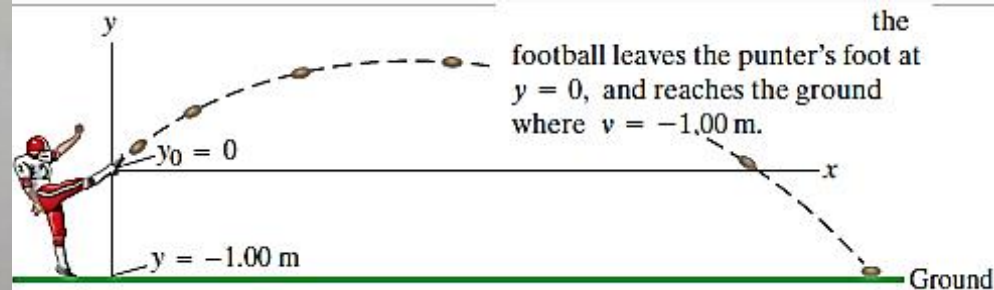
But $t = ??$

$$\Delta y = u_y t - \frac{1}{2} g t^2$$

$$(y - y_0) = u_y t - \frac{1}{2} g t^2$$

$$-1 = 12t - \frac{1}{2}(9.8)t^2$$

$$-1 = 12t - 4.9t^2$$



$$4.9t^2 - 12t - 1 = 0 \quad (ax^2 + bx + c = 0)$$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$t = \frac{-(-12) \pm \sqrt{(12)^2 - 4(4.9)(-1)}}{2(4.9)} = 2.53 \text{ and } -0.081$$

$$x = u_x t$$

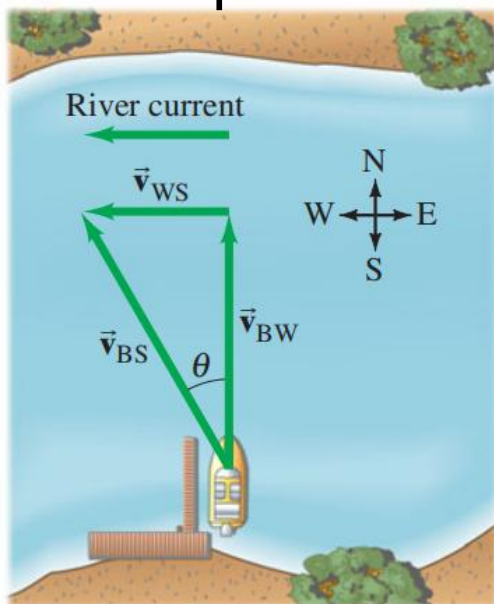
$$= (16)(2.53) = 40.5 \text{ m} \#$$

3-7 Relative Velocity

- We now consider how observations made in different frames of reference are related to each other.
- It is important, to draw a diagram and use a careful labeling process.
- Each velocity is labeled by two subscripts:

“The first refers to the object, the second to the reference frame in which it has this velocity.”

Example:



$\vec{v}_{BS}$ = velocity of **B**oat with respect to the **S**hore,

$\vec{v}_{BW}$ = velocity of **B**oat with respect to the **W**ater,

$\vec{v}_{WS}$ = velocity of **W**ater with respect to the **S**hore (river current).

By writing the subscripts using this convention, we see that the inner subscripts (the two **W**'s) on the right-hand side are the same as the two subscripts for the sum vector on the left,

the outer subscripts on the right (the **B** and the **S**) are the same as the two subscripts for the sum vector on the left,

3-7 Relative Velocity

- The equation can be extended to three or more velocities.
- For example, if a fisherman on the boat walks with a velocity relative to the boat, his velocity relative to the shore is

A ~~$\vec{v}_{FS} = \vec{v}_{FB} + \vec{v}_{WS} + \vec{v}_{BW}$~~

Which one is correct?

B $\vec{v}_{FS} = \vec{v}_{FB} + \vec{v}_{BW} + \vec{v}_{WS}$

A or B

The equations involving relative velocity will be correct when:

- adjacent inner subscripts are identical and
- the outermost ones correspond exactly to the two on the velocity on the left of the equation.

!! But this works only with plus signs (on the right), not

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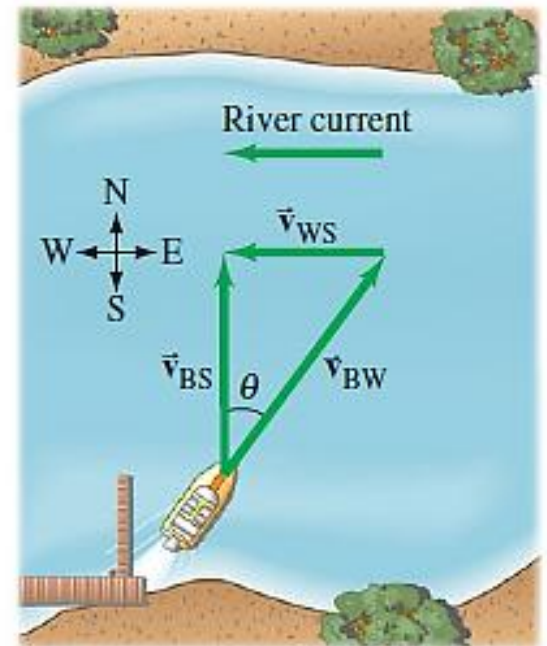
Heading upstream. A boat's speed in still water is $v_{BW} = 1.85 \text{ m/s}$. If the boat is to travel north directly across a river whose westward current has speed $v_{WS} = 1.20 \text{ m/s}$, at what upstream angle must the boat head? (See Fig. 3–29.)

If the boat heads straight across the river, the current will drag the boat downstream (westward). To overcome the river's current, the boat must have an upstream (eastward) component of velocity as well as a cross-stream (northward) component. Figure 3–29 has been drawn with $\vec{v}_{BS}$, the velocity of the **B**oat relative to the **S**hore, pointing directly across the river because this is where the boat is supposed to go. (Note that $\vec{v}_{BS} = \vec{v}_{BW} + \vec{v}_{WS}$.)

Vector $\vec{v}_{BW}$ points upstream at angle θ as shown.

$$\sin \theta = \frac{v_{WS}}{v_{BW}} = \frac{1.20 \text{ m/s}}{1.85 \text{ m/s}} = 0.6486.$$

$$\theta = 40.4^\circ$$



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Heading across the river. The same boat ($v_{BW} = 1.85 \text{ m/s}$) now heads directly across the river whose current is still 1.20 m/s . (a) What is the velocity (magnitude and direction) of the boat relative to the shore? (b) If the river is 110 m wide, how long will it take to cross and how far downstream will the boat be then?

The boat now heads directly across the river and is pulled downstream by the current, as shown in Fig

The boat's velocity with respect to the shore, $\vec{v}_{BS}$, is the sum of its velocity with respect to the water, $\vec{v}_{BW}$, plus velocity of the water with respect to the shore, $\vec{v}_{WS}$:

$$\vec{v}_{BS} = \vec{v}_{BW} + \vec{v}_{WS}.$$

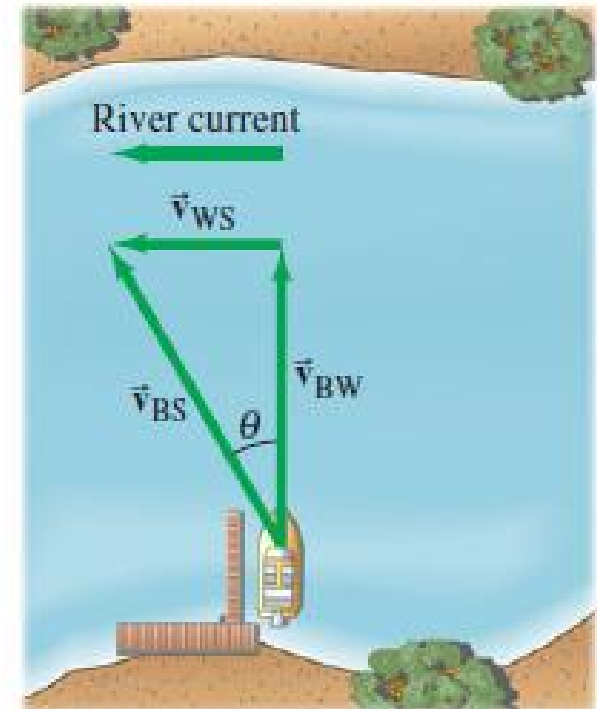
(a) Since $\vec{v}_{BW}$ is perpendicular to $\vec{v}_{WS}$ we can get $\vec{v}_{BS}$ using the theorem of Pythagoras:

$$v_{BS} = \sqrt{v_{BW}^2 + v_{WS}^2} = \sqrt{(1.85 \text{ m/s})^2 + (1.20 \text{ m/s})^2} = 2.21 \text{ m/s}$$

$$\tan \theta = v_{WS}/v_{BW} = (1.2 \text{ m/s})/(1.85 \text{ m/s}) = 0.6486$$

$$\theta = 33.0^\circ$$

A boat heading directly across a river whose current moves at 1.20 m/s .



Heading across the river. The same boat ($v_{BW} = 1.85 \text{ m/s}$) now heads directly across the river whose current is still 1.20 m/s . (a) What is the velocity (magnitude and direction) of the boat relative to the shore? (b) If the river is 110 m wide, how long will it take to cross and how far downstream will the boat be then?

$$b) V = \frac{D}{t}$$

Travel time = time the boat takes to cross the river.

$$D = 110 \text{ m}$$

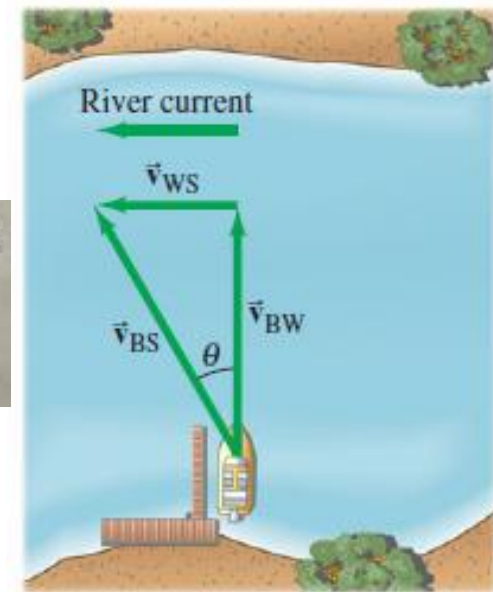
$$\begin{aligned} \therefore t &= \frac{D}{v} \\ &= \frac{D}{v_{BW}} \\ &= \frac{110}{1.85} \\ &= 59.5 \text{ s} \end{aligned}$$

How far down stream will the boat be ??

$$\begin{aligned} d &= v_{ws} t \\ &= (1.2)(59.5) \\ &= 71.4 \text{ m} \end{aligned}$$

→ The boat will have been carried downstream, in this time (59.5s)

A boat heading directly across a river whose current moves at 1.20 m/s .



$$\tan \theta = \frac{d}{D}$$

$$d = D \tan \theta$$

$$d = (110 \text{ m}) \tan(33^\circ)$$

$$d = 71.4 \text{ m}$$

Till now, you supposedly know how to:

- Define Scalar And Vector Quantities.
- Perform Addition & Subtraction Of Vectors Using Graphical Method & Component Method.
- Resolve Vector Into Two Perpendicular Components (X and Y Axes).
- State The Properties In The Two-dimensional Motion Of Projectile Motion
- Solve Problems On Projectile Motion by applying Kinematics Equation
- Express Relative Velocity Of Object With Respect To A Reference Frame



- Please scan for the attendance record